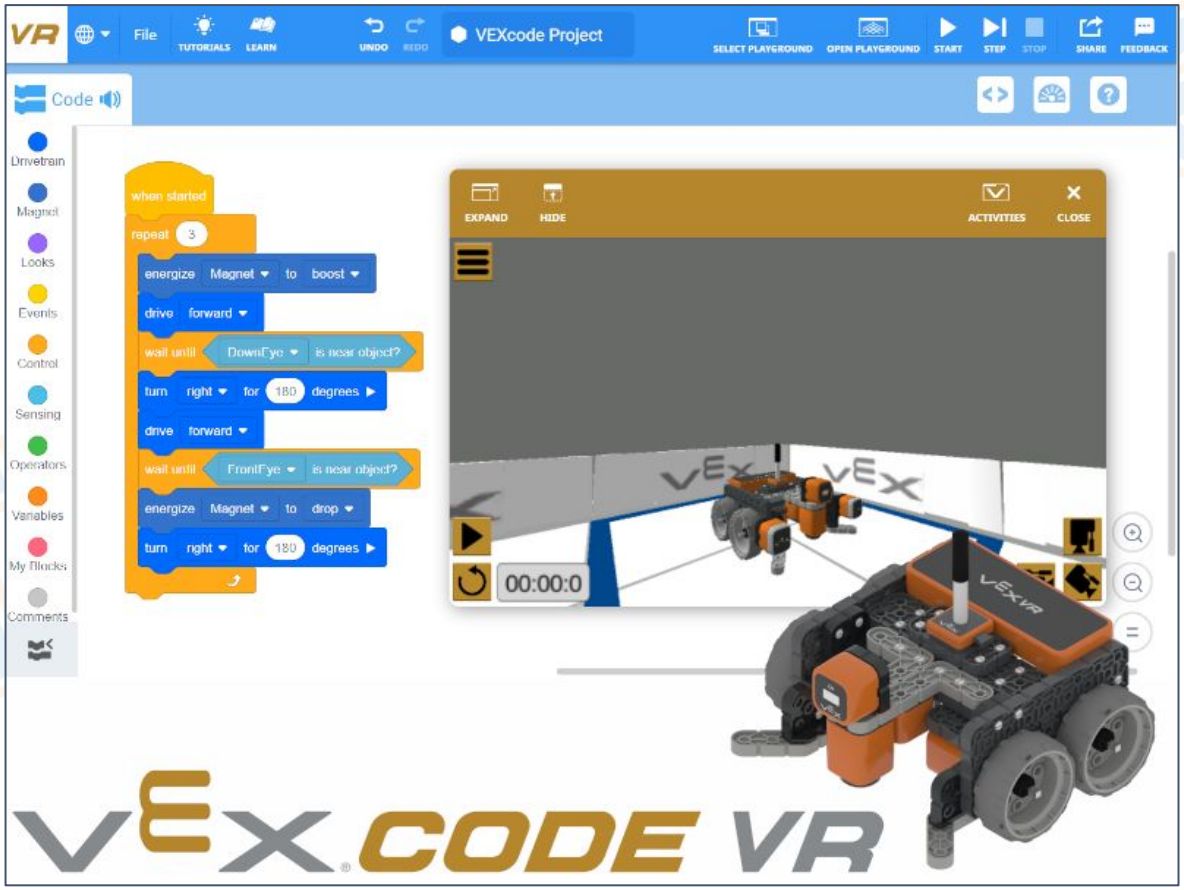
Introduction to VR: Vex Code!





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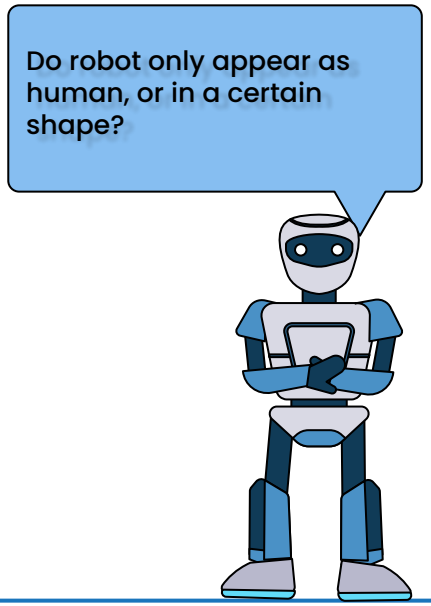
VR: Vex Code Outline

- Robotics, how do they work?
- Into the Block Coding World
- Introduction to Virtual Robots(VR) : Vex Code
- Castle Crasher Challenge
- Autonomous Sweeper Equipment
- The Smart Mouse



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Robotics, How do they work?



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Lets Learn what type of robot are there!

Mainly robot has 3 type divided among the all the type :

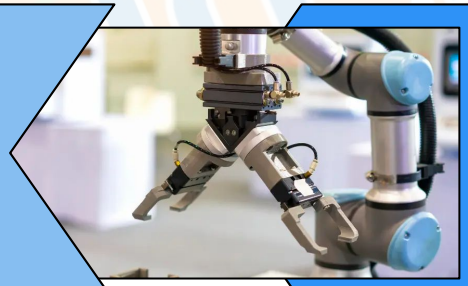


Industrial Robot

- Industrial robots automate tasks with enormous precision.
- AI integration enables predictive maintenance and adaptive responses.
- Flexible robotic solutions adapt quickly to changing production lines.

Cobots

- Cobots streamline workflows through human-robot collaboration.
- They assist technicians with delicate assembly tasks.
- Production lines integrate manual workstations into automation.



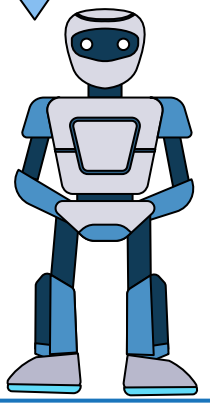


Mobile Robots

- Cobots streamline workflows through human-robot collaboration.
- They assist technicians with delicate assembly tasks.
- Production lines integrate manual workstations into automation.

This is the standard in our industry using are known.

Do you know if there is more type of robot our generation are using?

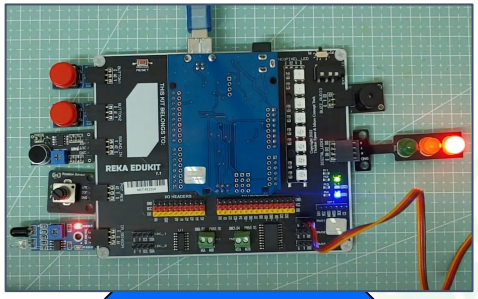


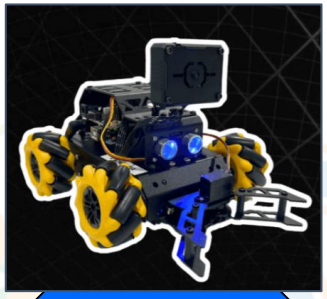
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What about the other type?

With Specific use and situation we have approach with type of :



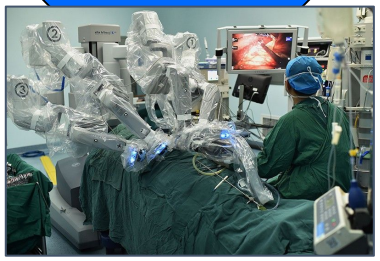




Humanoid Robots Mimic humans for intuitive assistance.	**Educational Robots** Facilitate interactive, hands-on STEM learning.	**Autonomous Mobile Robots (AMR)** AMRs navigate dynamically without fixed infrastructure.
Nanobot Enable precise treatments at the cellular level.	**Services Robots** Streamline tasks in hospitality and healthcare.	**Medical Robots** Enhance surgical precision and remote patient care.

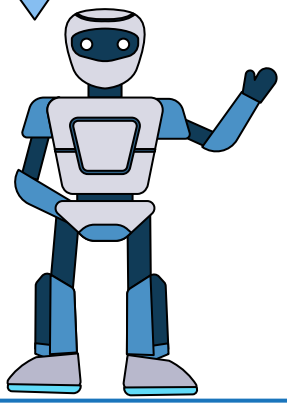






Can you guess my type of robot?

Maybe there will be much more type of robot in the future!



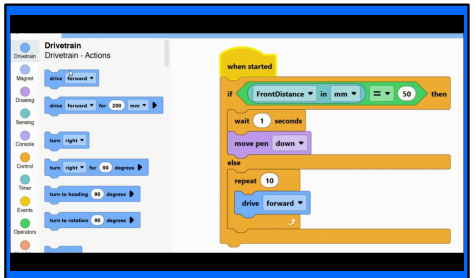
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How Education Robot contribute?

Every person involved in robot learn on the basic first!

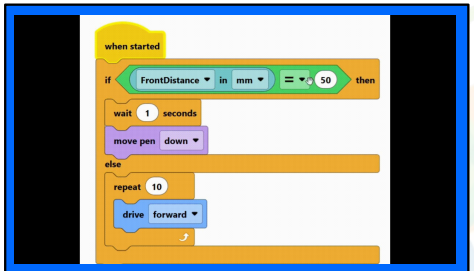
01 New Concept



To learn more abstract concepts like variables, loops, or physics on coding.

Education Robotics makes these invisible concepts physical for future tech use.

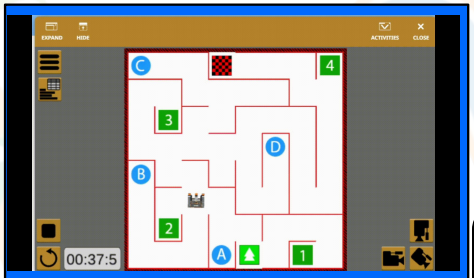
02 Computational Thinking



Get used to precision and logic thinking in coding for every single action.

Education Robotics train ourselves how to plan and make decision for each logic applied in our code.

03 Resilience

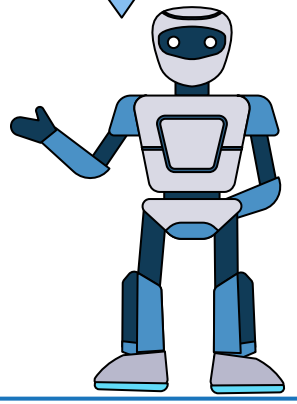


In robotics, failure is almost guaranteed. A robot rarely works perfect the first time.

Education Robotics builds persistence, essential on tackling complex Robotic challenges or development.

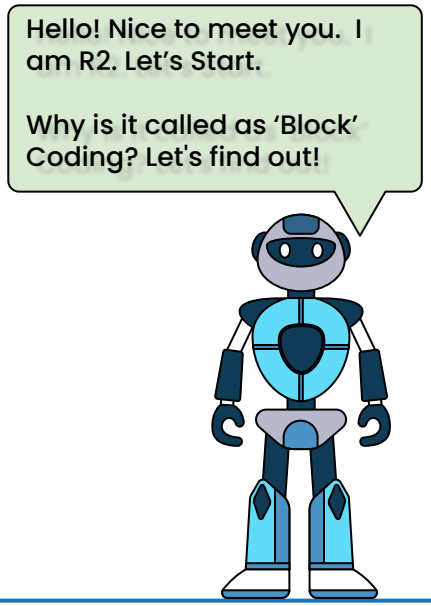
With Education robot, student can learn early up to 8 years old for preparation on advance work on robotics.

Now, are you ready to learn actual coding?



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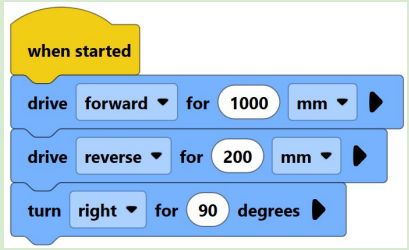
Into the Block Coding World



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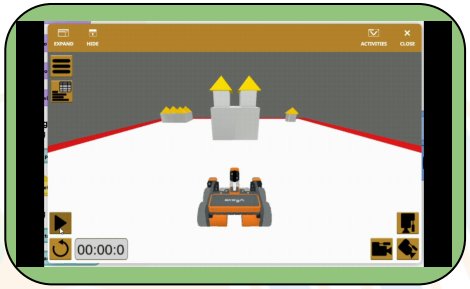
Event Starter

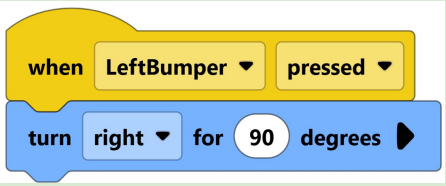
Every single code need someone to make them start!



When Started

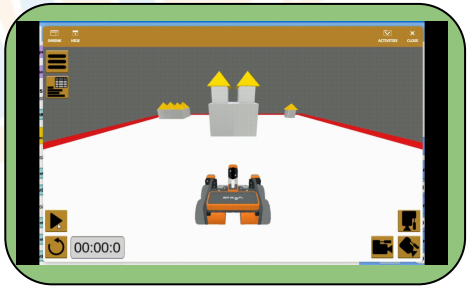
Simply pressing play will execute the code below it





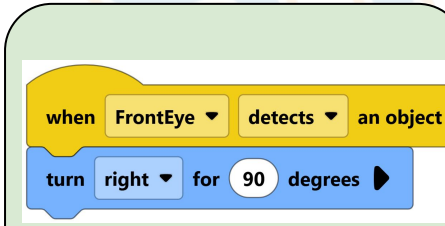
When LeftBumper Pressed

Execute the code below when the robot touch the block at its front.



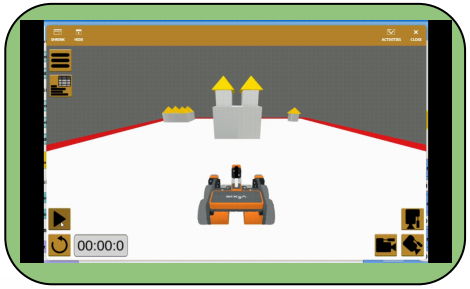
Can You see the difference?

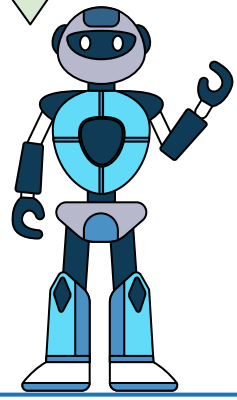
Depending on Event Start Condition, the car will turn based on its range within the block.



When FrontEye detect

Execute the code below when the robot sensor see block at its front



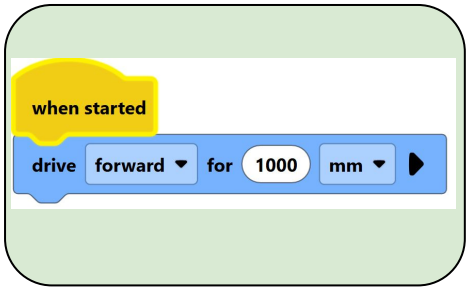


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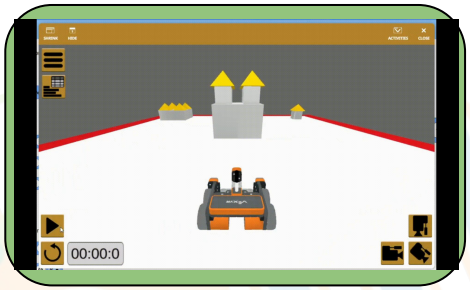
DriveTrain Action

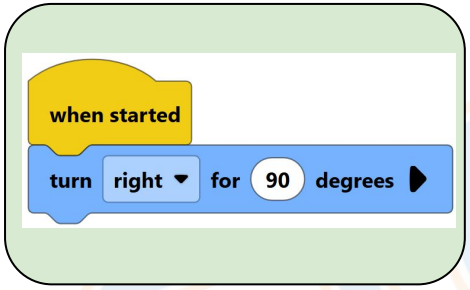
Now Lets Learn what the robot could do after start.



Drive Forward/Reverse

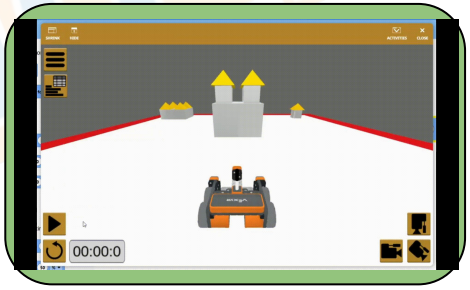
Make the robot move either continuously or according to range.





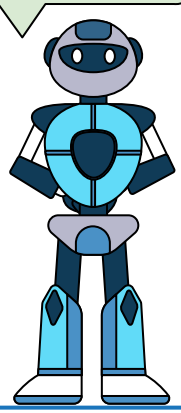
Turn Right/Left

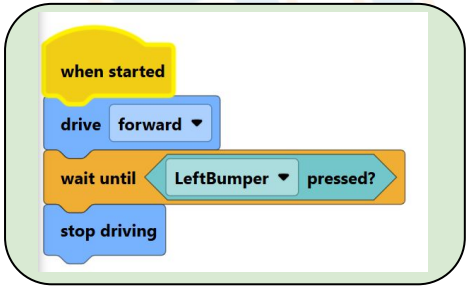
Make the robot Turn according to direction and within degrees input.



This is the fundamental action for robot to perform

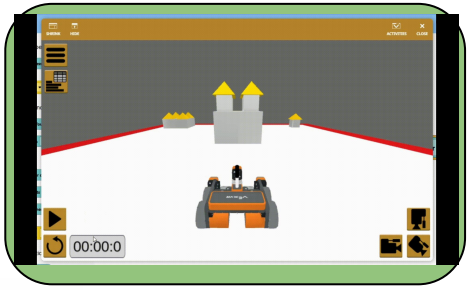
Make sure to use it well for accurate action!





Stop Driving

Useful on condition statement for stopping the car if condition success.

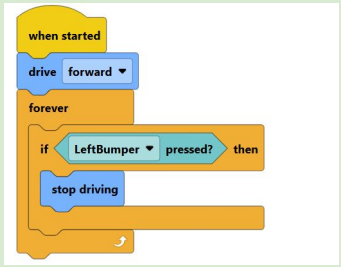


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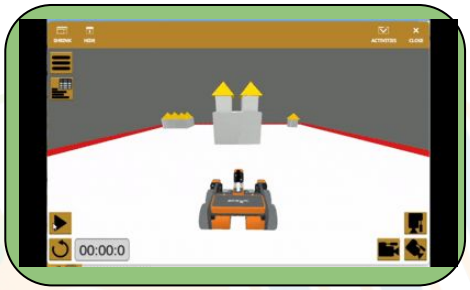
Sensing

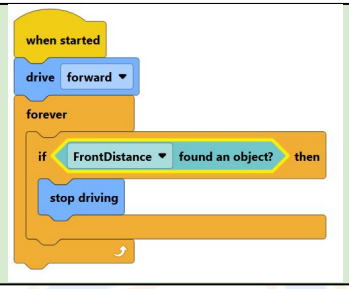
This is where the robot shine for multiple capabilities.



Left/Right Bumper Pressed

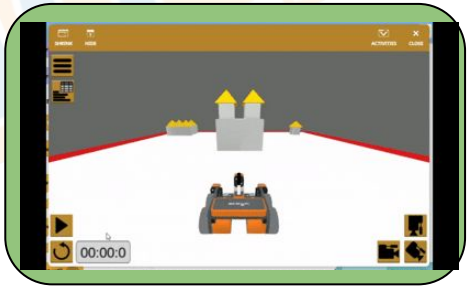
Sense the right or left bumper in front of robot on touch or press

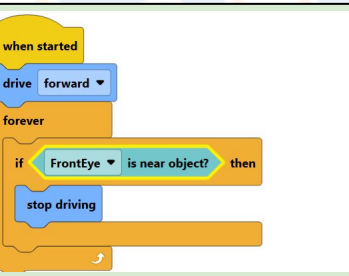




Front Distance found an object

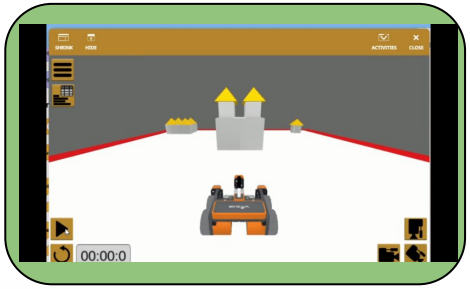
Sense in front of the robot if there is object front of it.





Front/Down Eye is near object

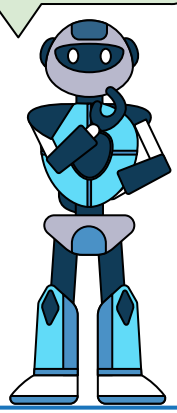
Sense in front of the robot if there is object near in front of it.



Sensing is a game changer for a robot.

It can interact with the surrounding with the robot for multi purpose.

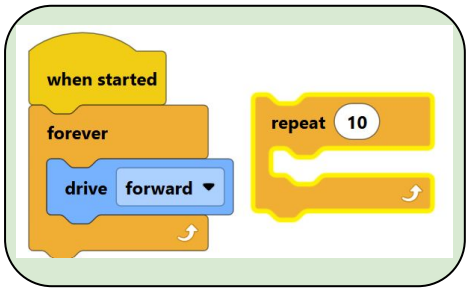
Any creative idea to use it for?



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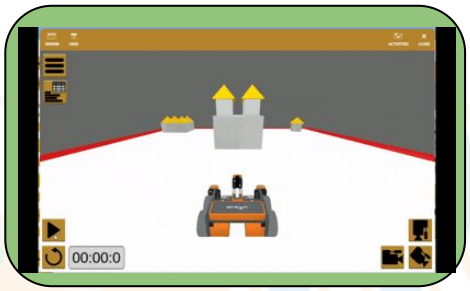
Controls

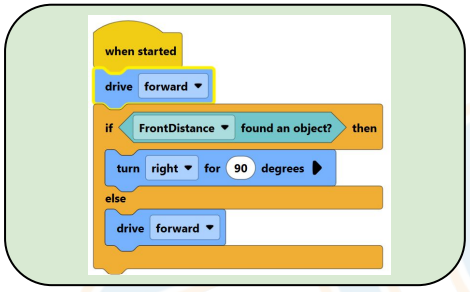
Here is where we can make a condition or repetition for our robots.



Forever/Repeat

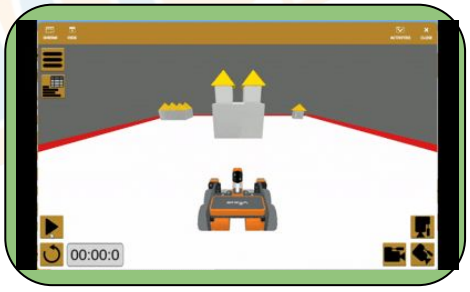
Make the code put inside executed either non stop or with value.





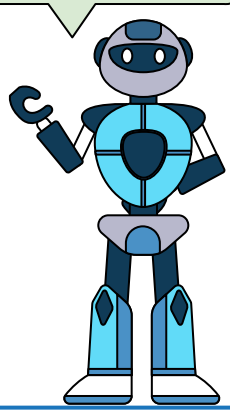
If Else

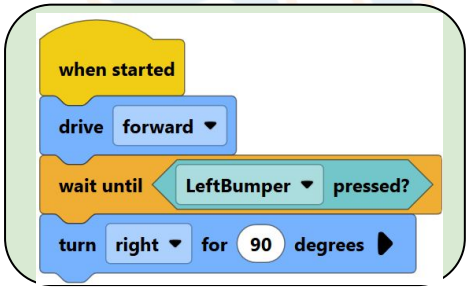
Enable Condition be put to ensure the robot will execute the code only if success Or it will execute Else.



Here is where we learn to make efficient code instead of copy paste same code.

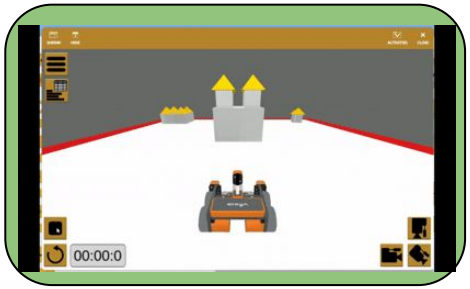
Also, with if else, we can get more accurate situation action.





Wait Until

It will keep execute the code on top of it until the condition success which it will execute code under it.

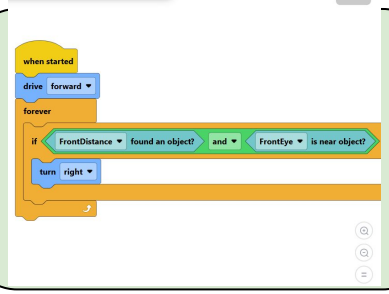


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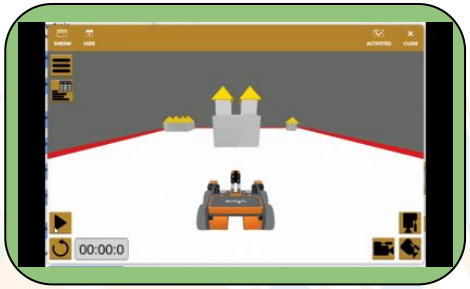
Operators

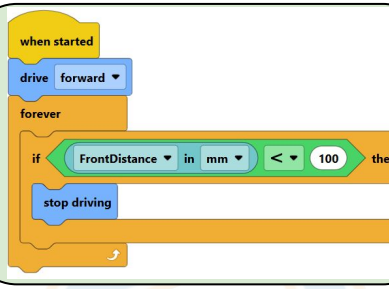
Operators can be help on condition statement for more detail and acc



And/Or

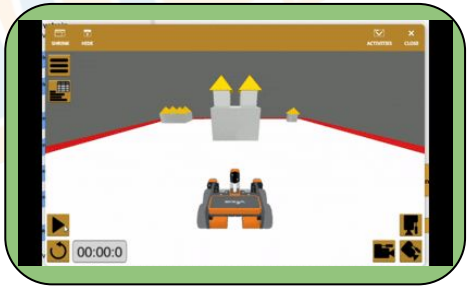
Make the robot move either continuously or according to range.

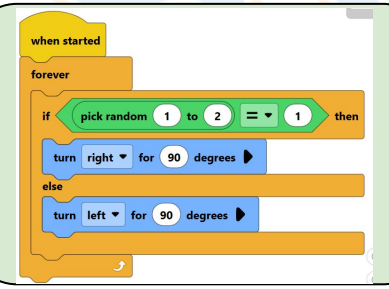




Less/More Than/Equal

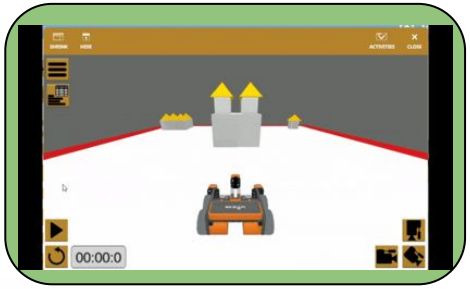
Make the robot Turn according to direction and within degrees input.





Pick Random

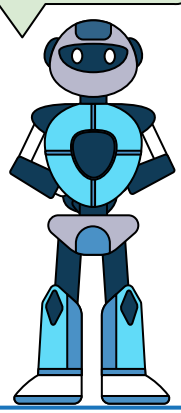
Useful on condition statement for stopping the car if condition success.



Operations usually is situational use case.

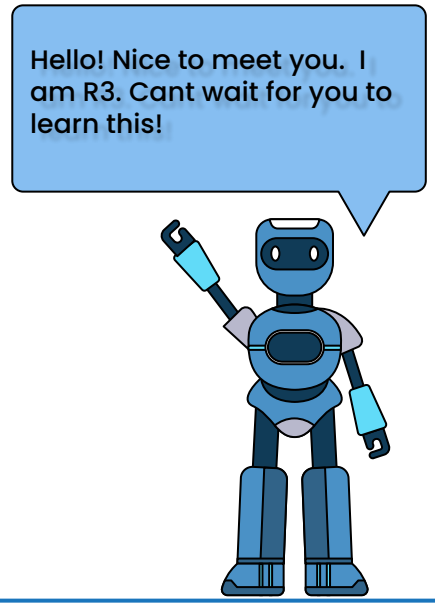
You rarely use them but become really helpful once the situation arise.

Make sure to learn them well!



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Introduction to Virtual Robot (VR): Vex Code

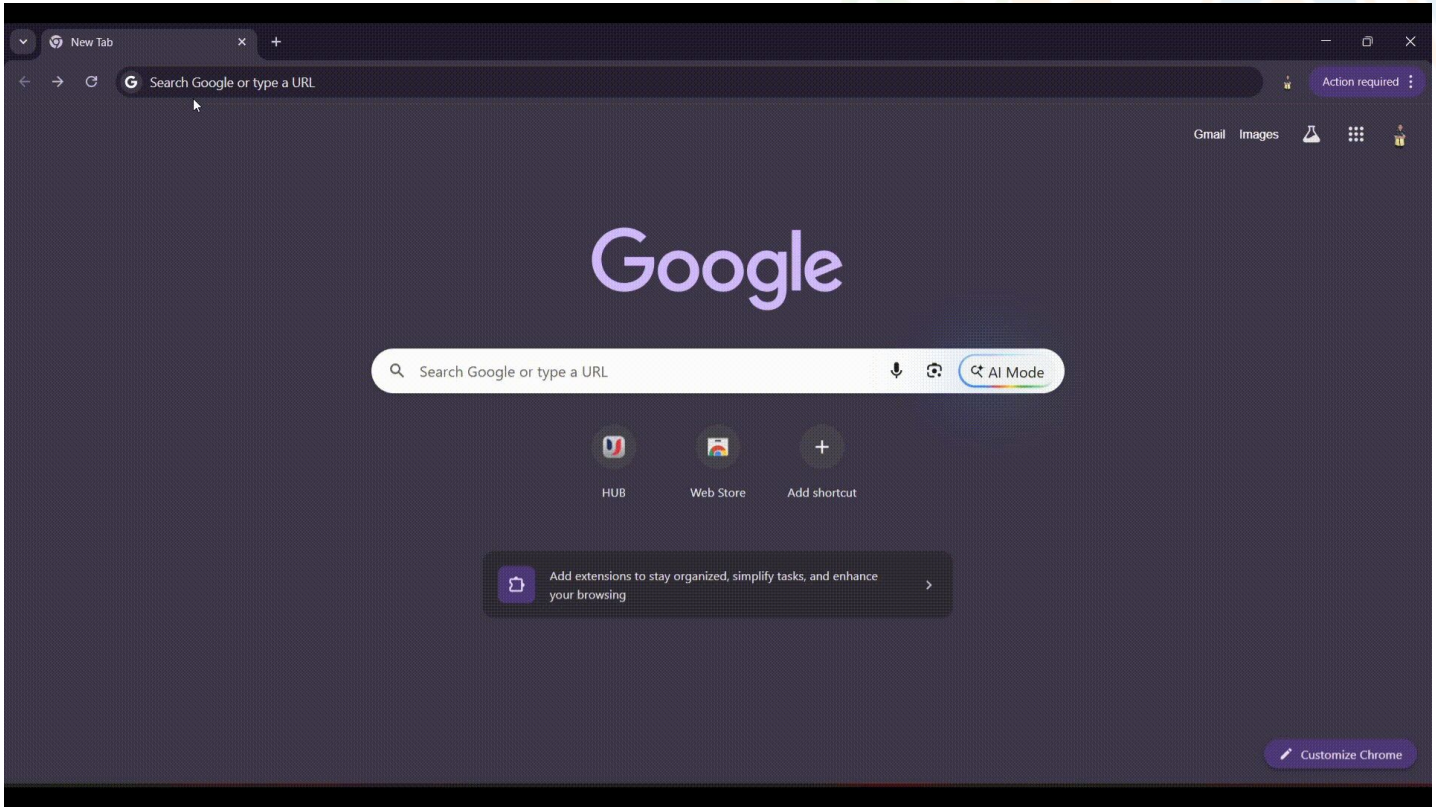


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Onto the web

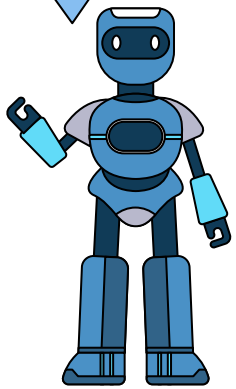
Let's open the website!

1) Type VR Vexcode in the browser
2) Make sure to choose the Top one looking like this!



Please Ask question if you do not understand!

Remember, asking is better than silence in unknown.



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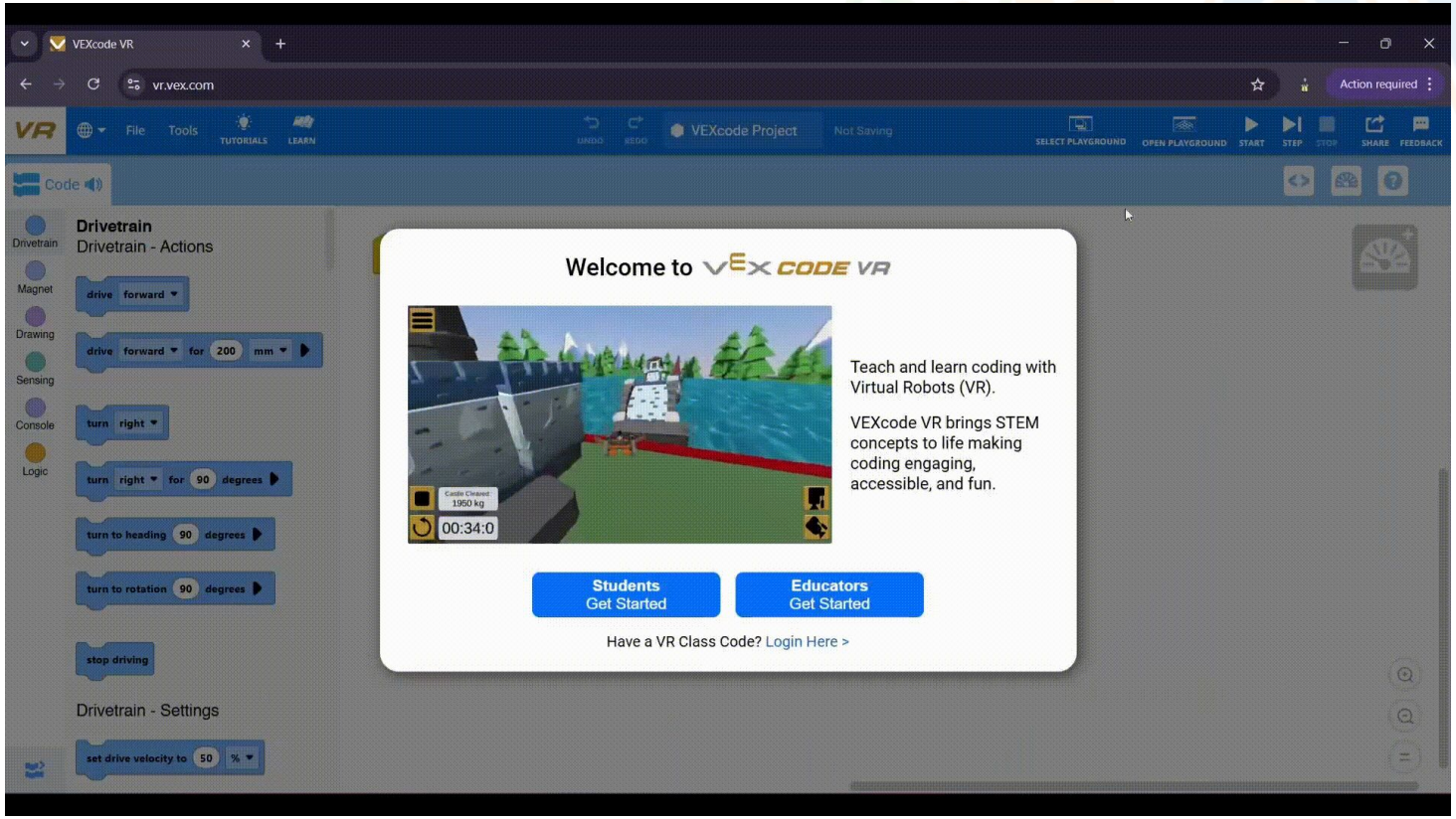
MICRO CONCEPT TECH

Dive Into the world of Robotic Coding

Time to learn what is the website capable of!

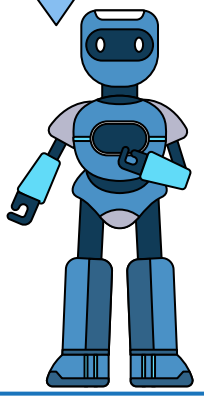
3) Now press ‘Students Get Started’

4) Follow the tutorial and read the description one by one.



VR Vexcode give short tutorial for you to understand how it works!

If you get lost, do not hesitate to take your time on understanding. Better late than never!.



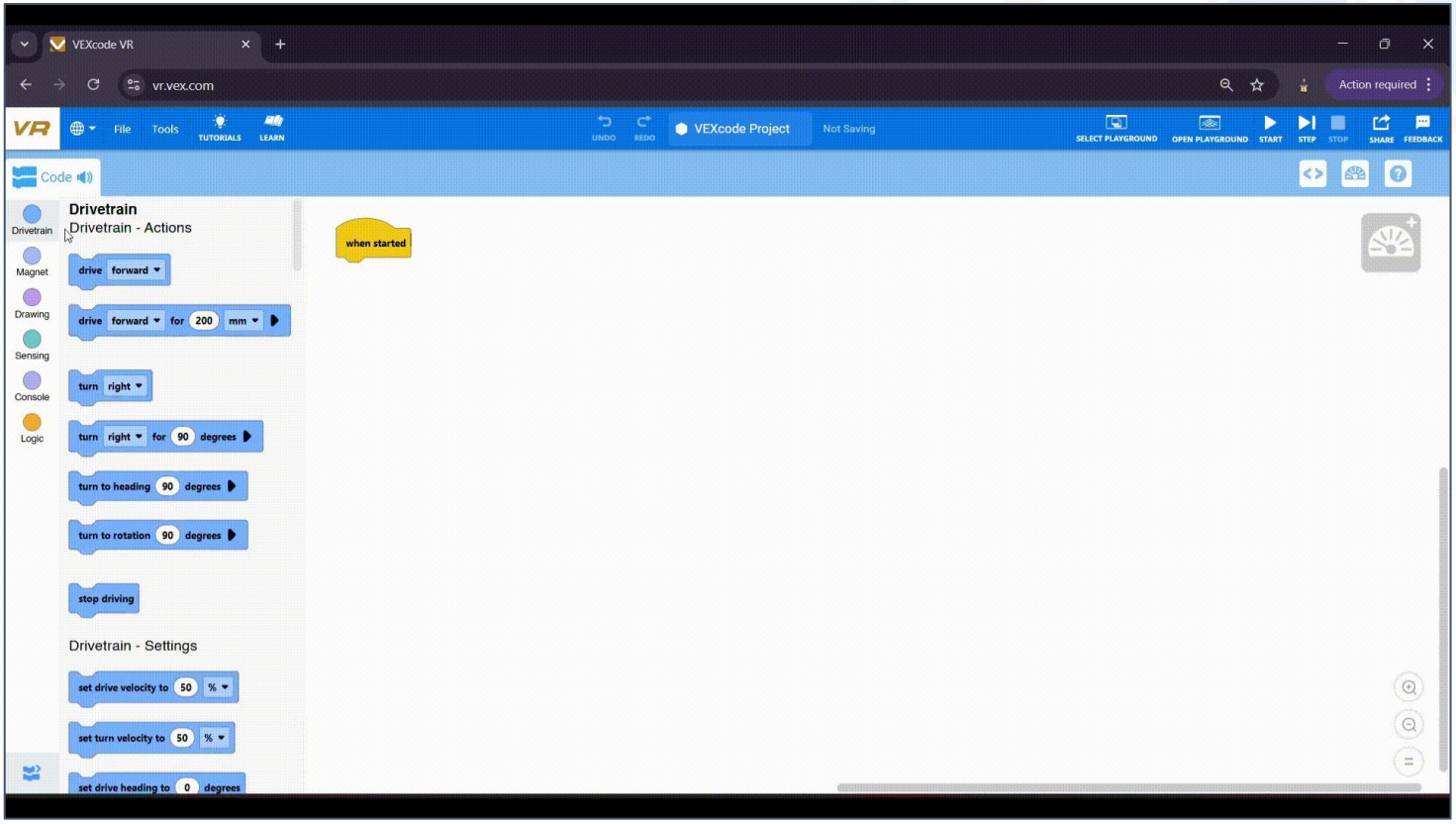
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The Drivetrain

Many code to choose from!

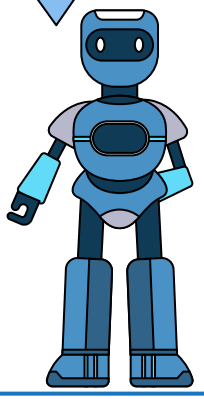
- Only basic are shown for the starter for ease of search, but you can expand the list to see much morefor the code choices.



We learn the basic from past slide.

Now you can see there is much more version of it.

Take your time on understanding it.

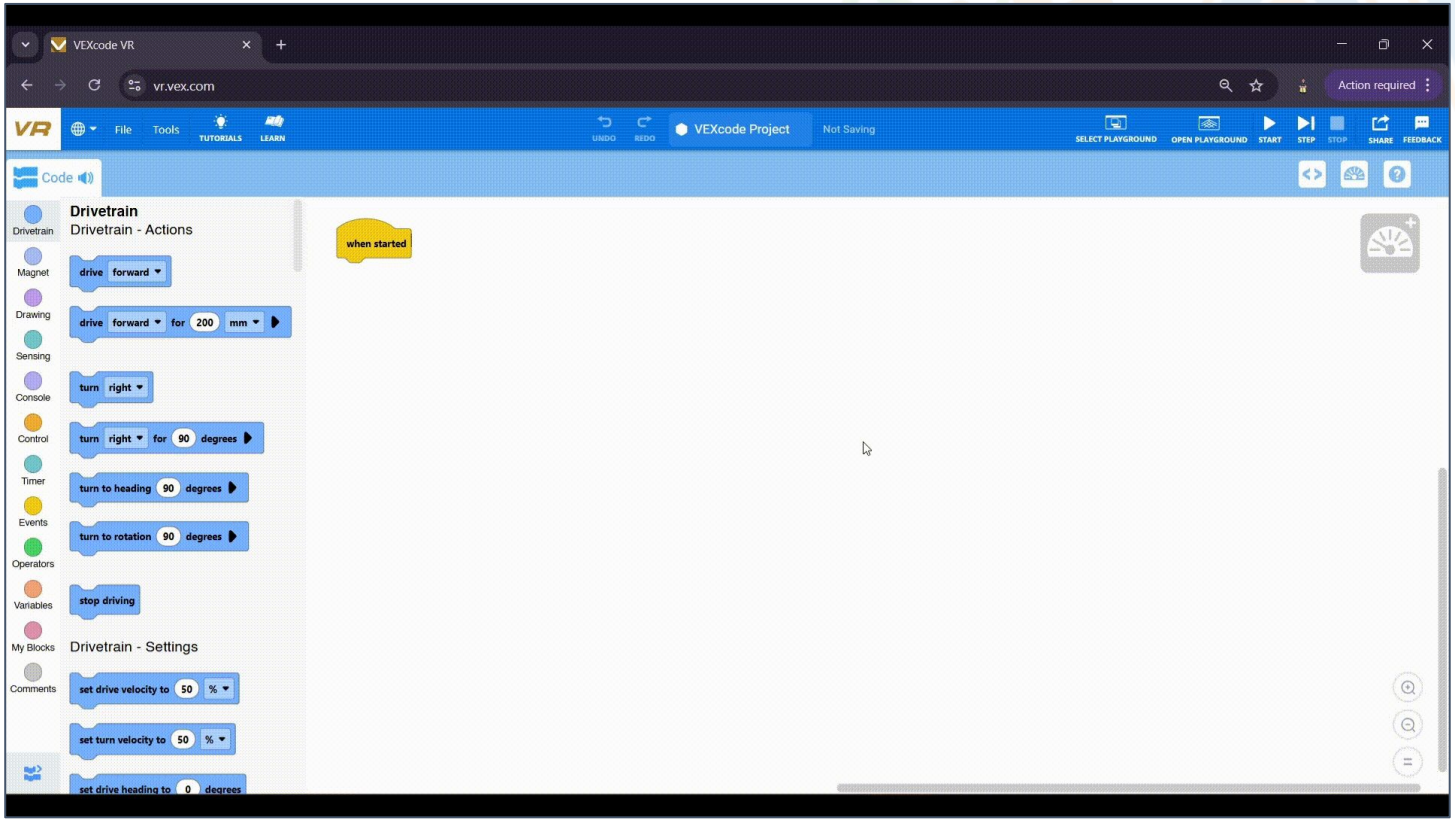


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The Virtualization of Robot

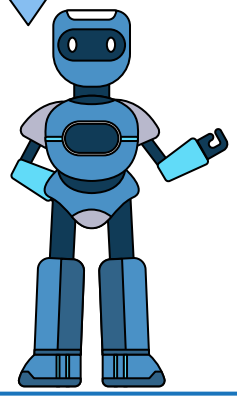
Here is where we make the robot play around the map.

- Choose from multiple of playground, but only for basics.
- The basics are enough for you to understand on multiple robot behaviour.



VR playgrounds is the game changer for you to learn

Instead of theory, now it's practical to learn.



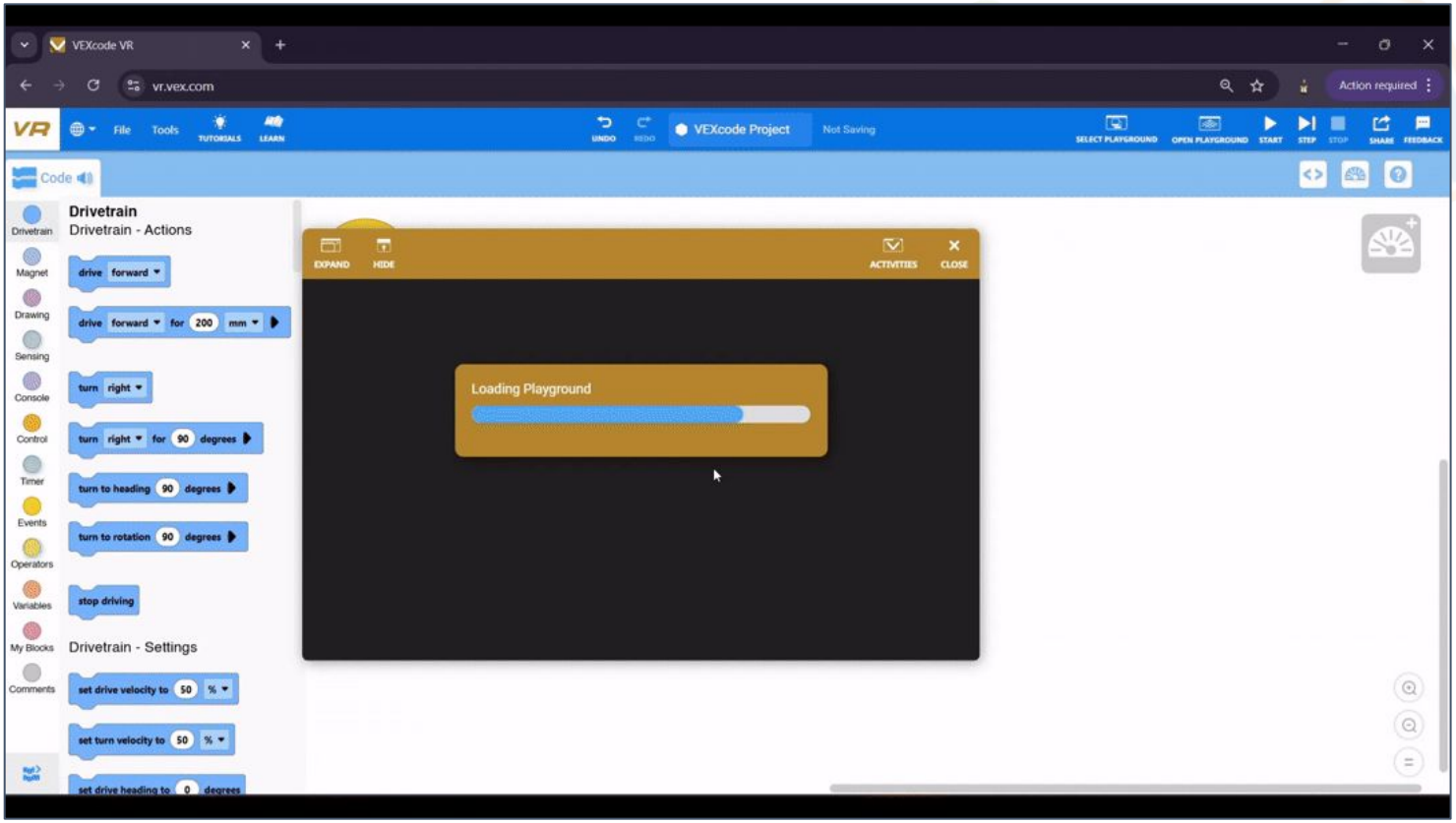
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Starting the Simulation

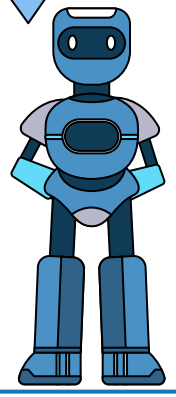
Now we can see how the robot simulation work

- Playground can be simulate and see in various function.
- You can see sensor, multiple camera and size from playground.



With the playground, its really exciting to see the robot perform with our code execution.

Faster method for us to see how our code performance on the robot.

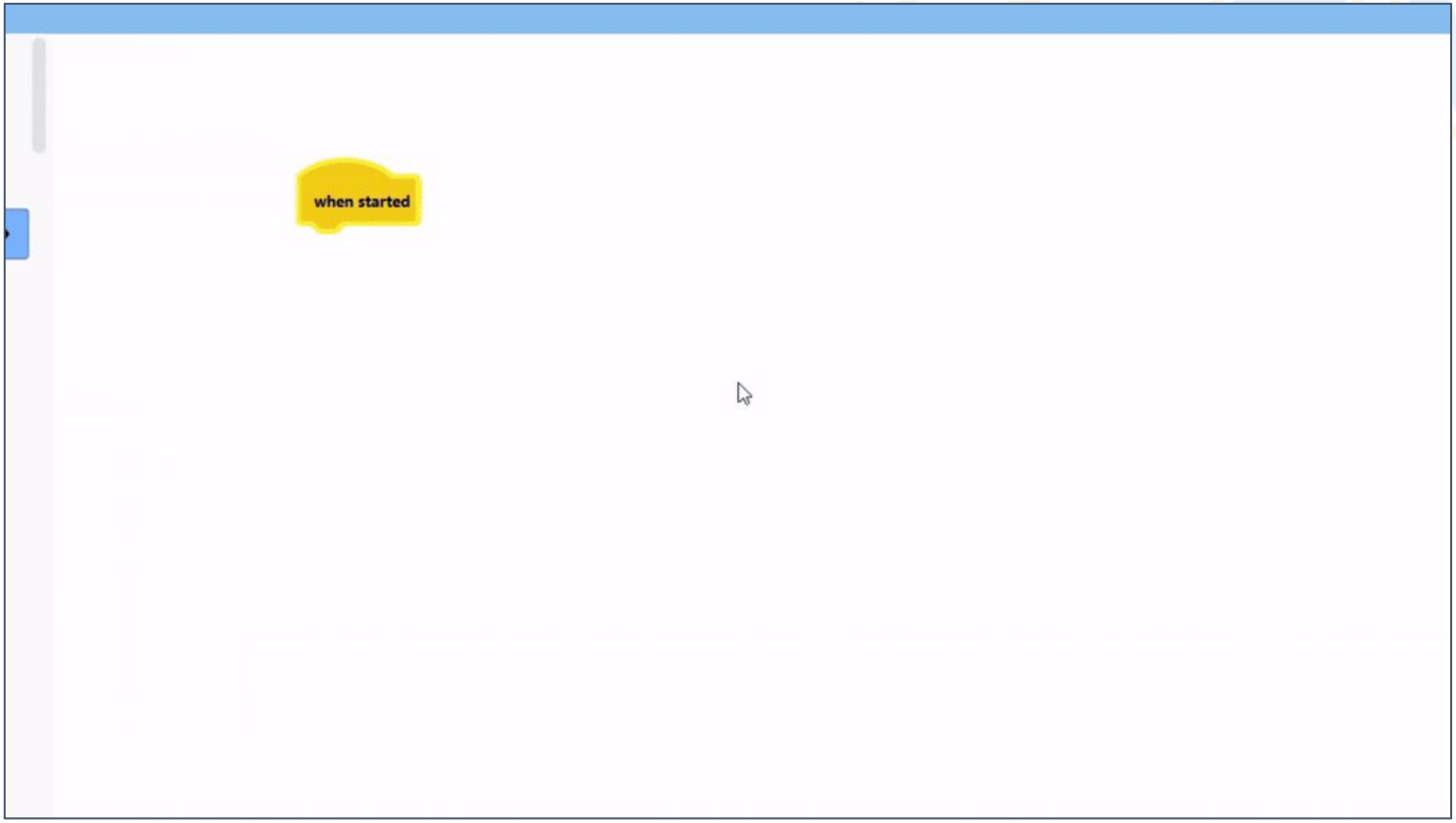


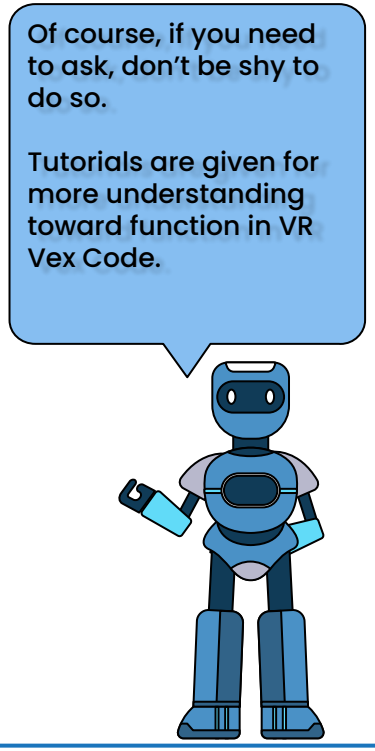
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Extra Tutorial

More Tutorial to learn if you are interested

- On Starting Vex Code for the first time, you already see the first tutorial.
- Now if you are still in lost, Vex Code VR give much more tutorial to learn.



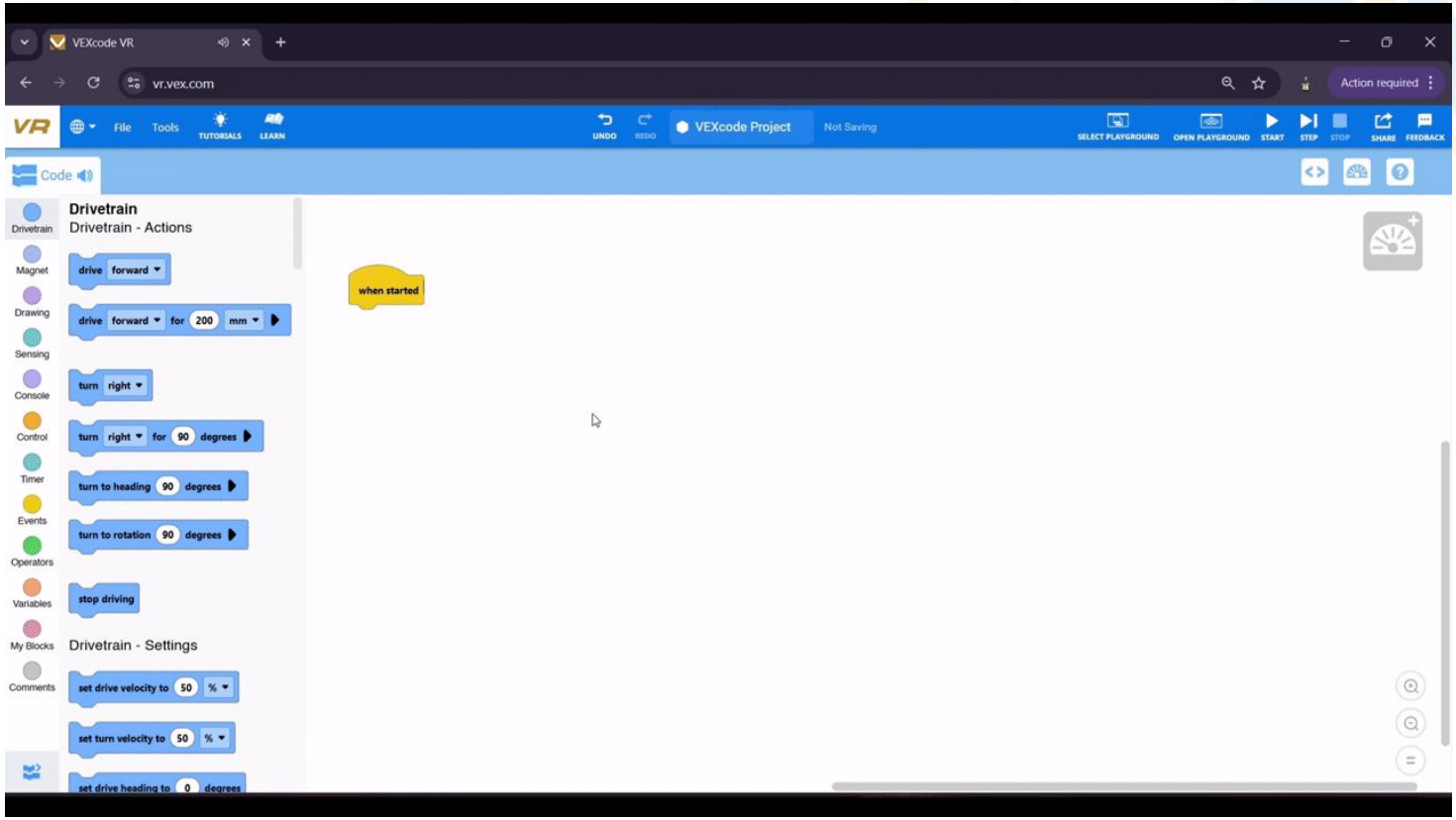


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Extra Material

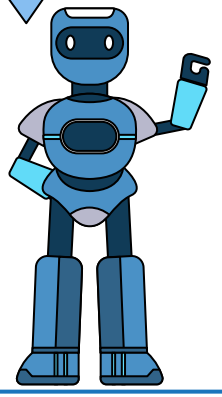
Learning advance function for the robot and feature in Vex Code!

- Learning Material provided let you learn more on advance function.
- It give Code instantly for example of how it's execute.



Extra Material are perfect for student who want to be advance.

Some of it might locked but many learning material can be access.

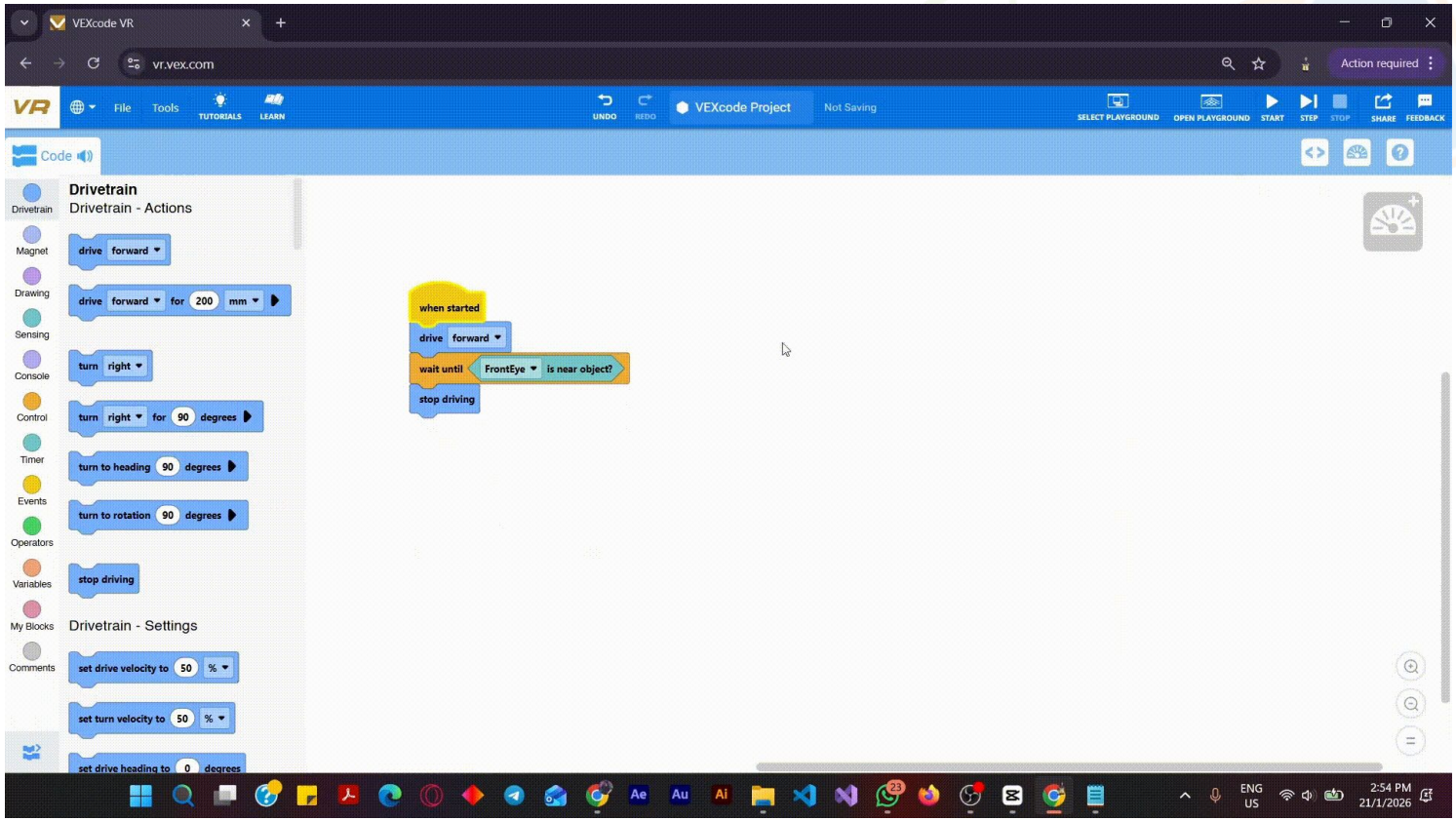


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Sensor Monitoring

As a part of robot, sensor play huge role on performing better.

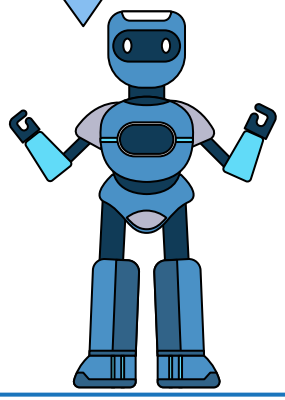
- Vex Code Vr provide multiple sensor monitoring in playground and outside.
- With sensor, you can monitor your robot logic movement on sensor attached.



With sensor monitoring, you can also see if something wrong on your robot.

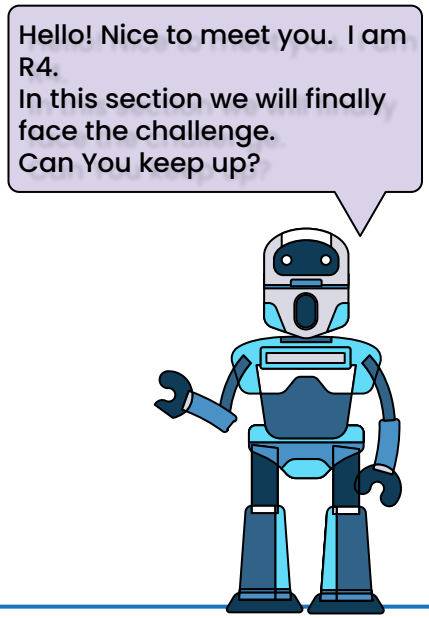
It Act as debugger, telling us what is going on with the sensors.

Really helpful for us.



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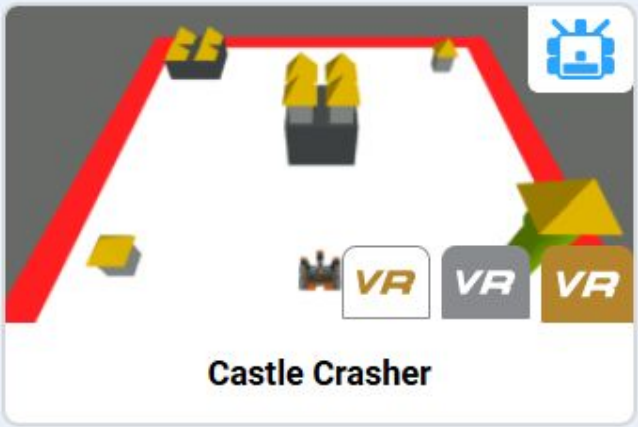
Castle Crasher Challenge



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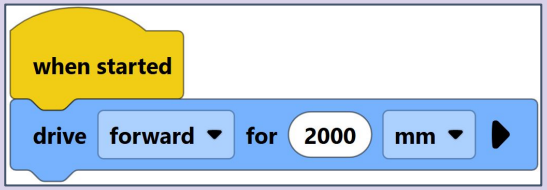
The First Trial

Let's try to knock One tower first.



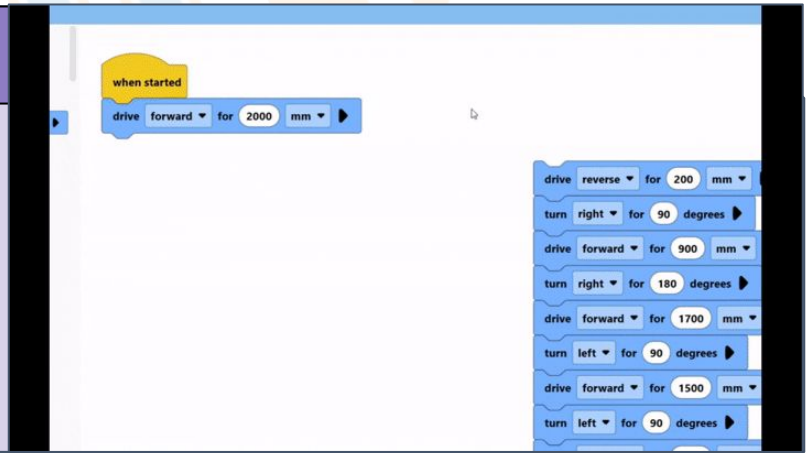
1 Setup on Castle Crasher

- Open Castle Crasher Playground
- Setup Code Same as below



Testing the Code 2

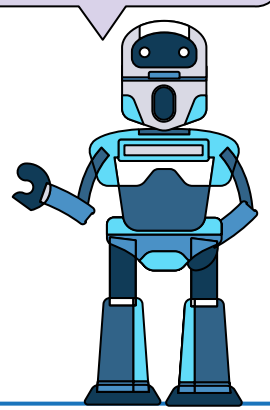
- Now, Try to run the code.
- See what happen?
- It can knock the castle but fall afterward.



We will start simple for understanding.

For our first it should be simple to execute.

Hope you can keep up!



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Debugging

Figuring out the problem and solution

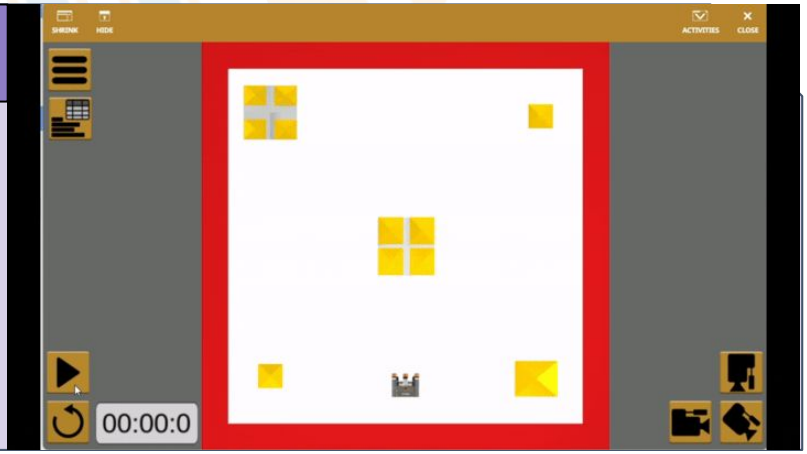
1 Adjusting The Code



- Now let's adjust the code a little bit.
- How bout we change the 'Drive Forward' value to 1700
- Can it stop at the right time?

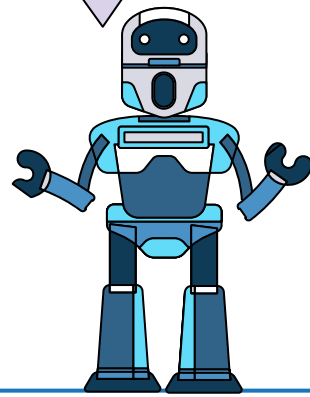
2 Testing the Code

- Now if you run the code, the robot no longer fall.
- It stop right before red line.



Debugging are a process where programmer figure out where its wrong and solution to fix it.

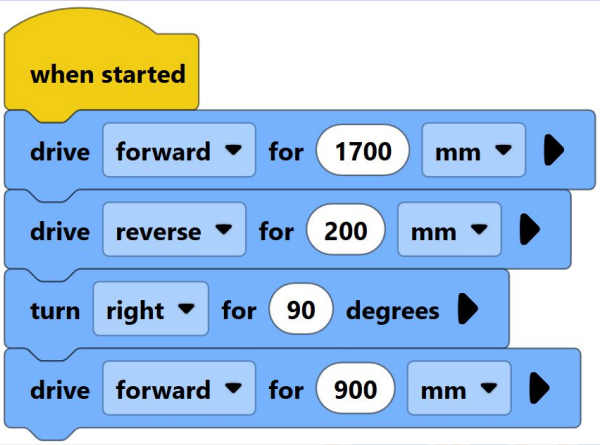
Fun Fact, if there is problem, we call it as 'bug'.



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Sequencing

Going to the other castle.



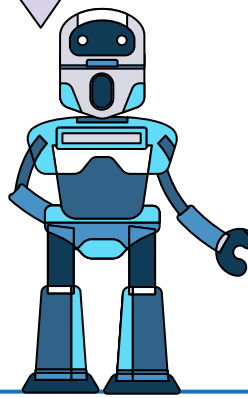
1 Additional Code

- Okay, now it don't fall we can add new code.
- Let's try to knock castle on the right.
- We can add Turn Right and Drive Forward with value as in image.

Can you still follow?

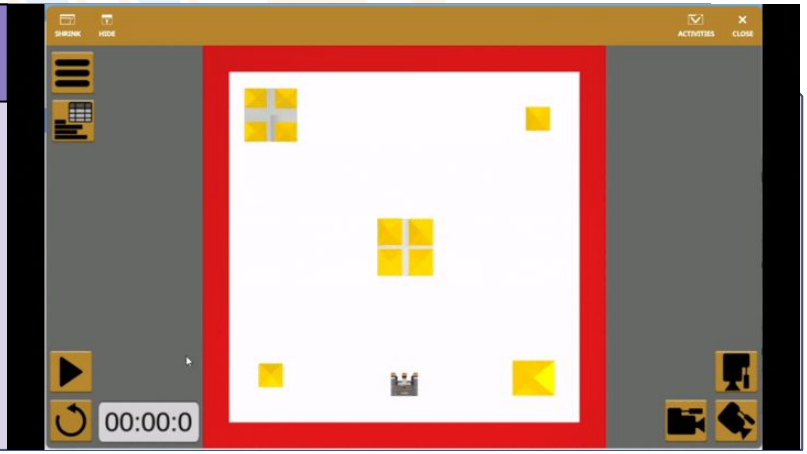
Let me know if anyone need help on understanding.

No one left behind!



Testing the Code 2

- The robot manage to hit the right castle.
- Now, can you figure out how to hit all the castle?

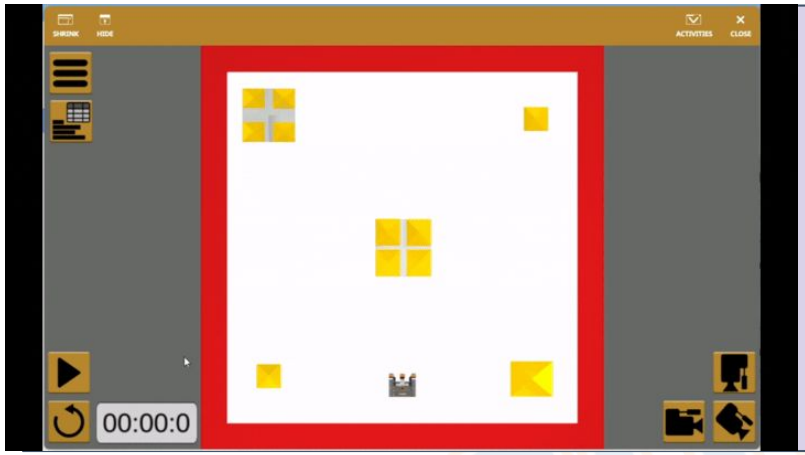


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Here Comes the Challenge!

Can you figure out how to knock/Move all the castle?

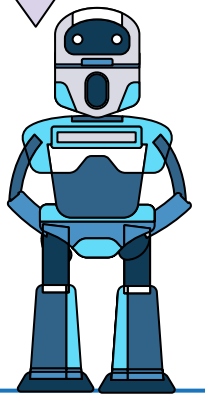


1 How to use the timer

- Okay, now it don't fall we can add new code.
- Let's try to knock castle on the right.
- We can add Turn Right and Drive Forward with value as in image.

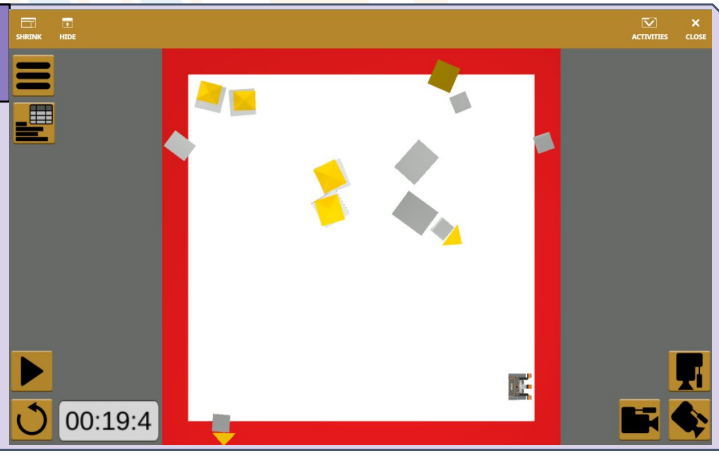
Now from what we learn before, hope you can complete this challenge.

Golden rule, achieve your goal first before making it efficient!



Send a Picture! 2

- Okay! You can try take a picture like in image on the right.
- Can you beat the time in the image?
- Don't be afraid to use other code too!
- All The Best!

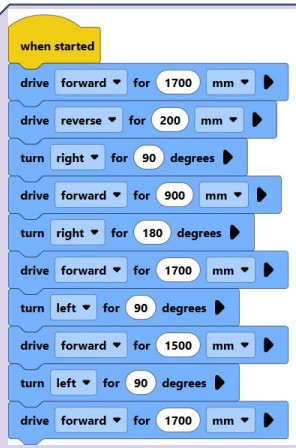


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The Answer

If you get the same result with different code, it's fine!

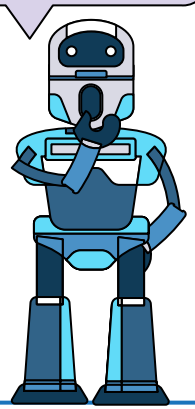


1 Subjective Answer

- In Coding, there is multiple answer that can lead to same result.
- If you manage to knock/move all the castle, then you are correct!
- Remember, there is always more efficient way with less code and faster timer.

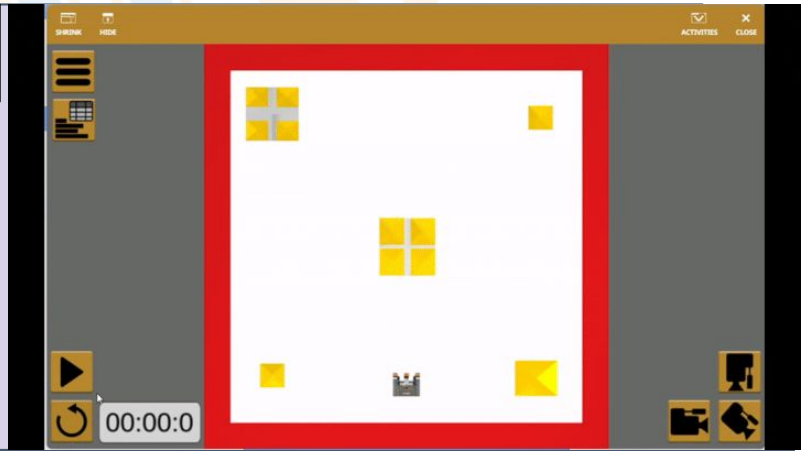
Anyone use different code execution?

Let me know!



Result from code 2

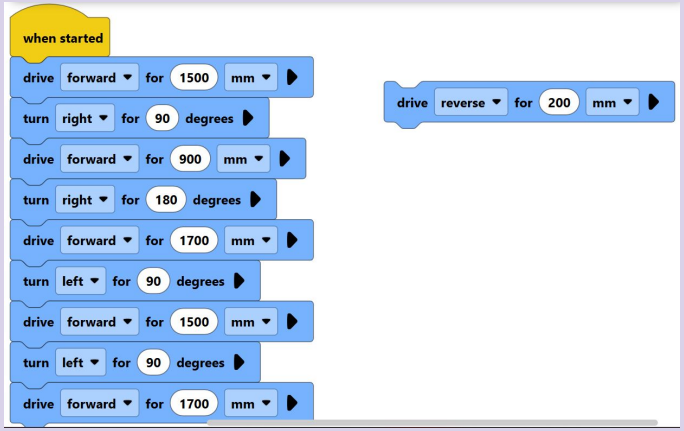
- Here, the robot manage to knock or at least move all the castle.
- So, do you beat the time?



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Conclusion

Congrats on finishing your first project!

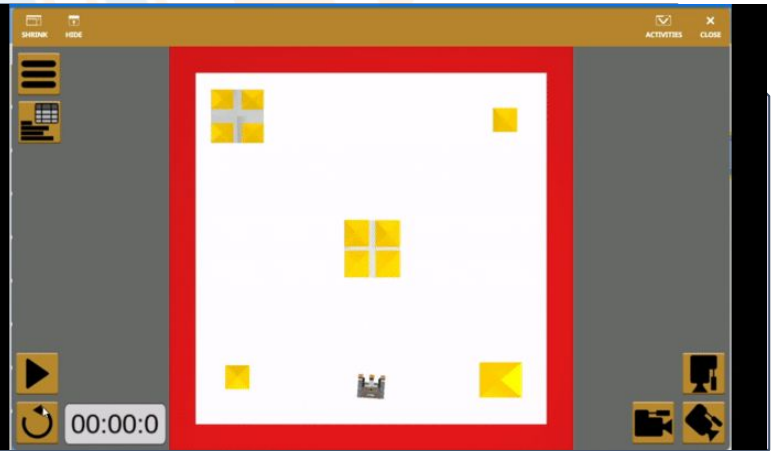


1 Coding Etiquette

- Remember, in coding, the less length of it, if it can reach the objective, it is always better.
- As per in image on the left, Removing 'drive reverse' and adjusting the 'drive forward' value to 1500 can achieve the same result.

Lesser Code Execution 2

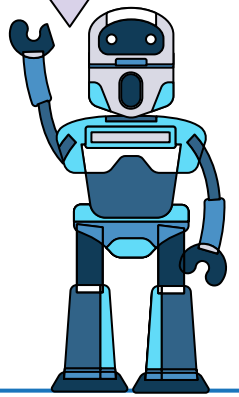
- As you can see, by removing 'drive reverse' and adjusting the 'drive forward' value, we can get faster timer by 1 second.
- 1 second might not seem a lot but imagine a more complex and multiple code execution.



WIth this, we finally finished the first project.

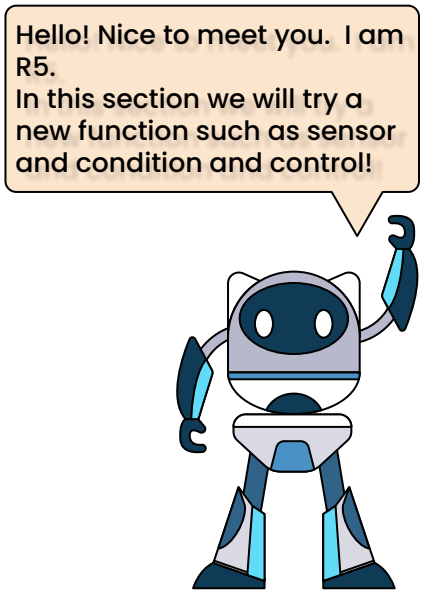
The next project will be more advanced.

Be prepared, coder Warrior!



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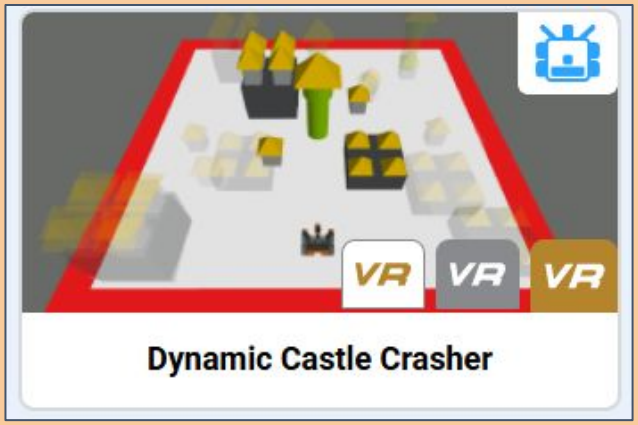
The Autonomous Sweeper

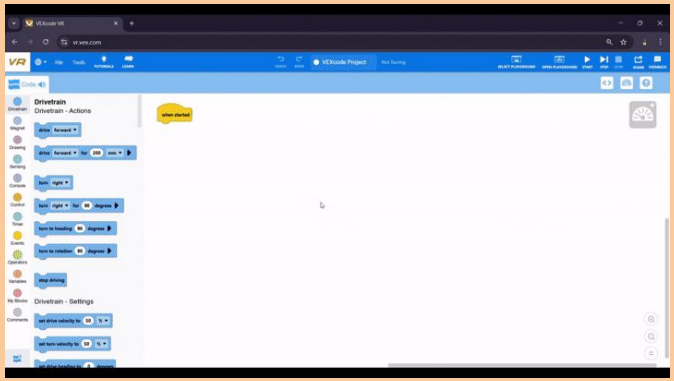


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The Dynamic Castle

Why is it called as dynamic castle?





1 Playground Setup

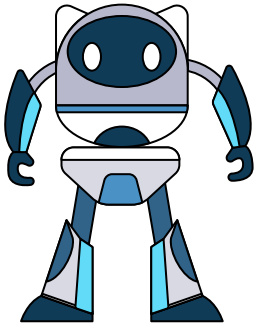
- Now, lets open the setup for dynamic castle crasher.
- What is the difference between the usual castle crasher and the dynamic one?

2 Steps to do

- Student can press playground first on top right of the screen.
- Then,scroll down and choose Dynamic Castle Crasher.
- As you can see, dynamic layout of castle change on every refresh!

Now instead of fixed position, we have randomized position of each castle.

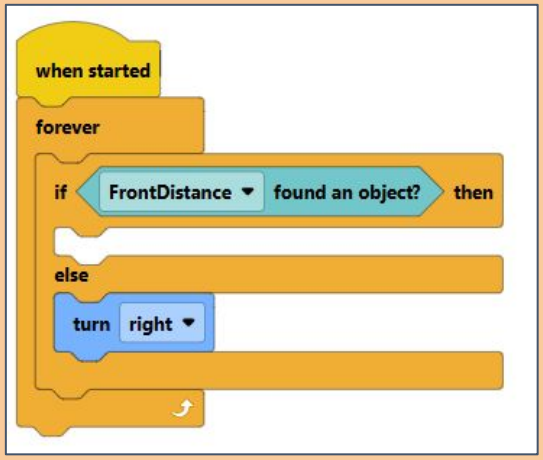
Let's figure out how can we tackle all the castle no matter where the castle is!

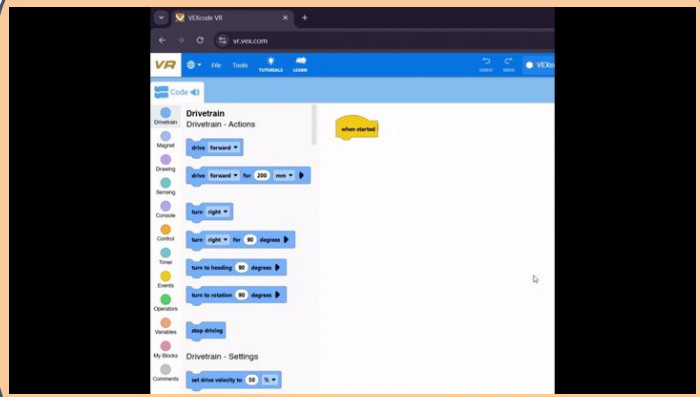


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Scanning Algorithm

Before any action, we need to detect the object first!





1 Scanning Algorithm

- Before we make the robot move, we need to make it able to scan its surrounding first.
- Block 'FrontDistance found an object?' is a good starting for scanning.
- We also want it to always scan with 'forever' block.

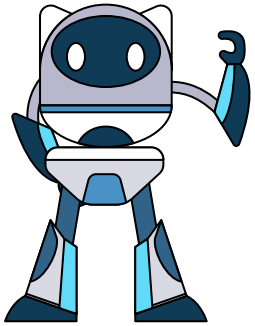
2 Code Setup

- Student can grab 'forever' block first and put it under when started.
- Inside 'forever', drag 'if else' and attach it inside.
- Then we can drag 'FrontDistance found an object?'and put it inside if.
- Finally, the turn right block in else.

The first step might take some time for several to follow.

But no worries and take your time.

Do not hesitate to ask any questions!

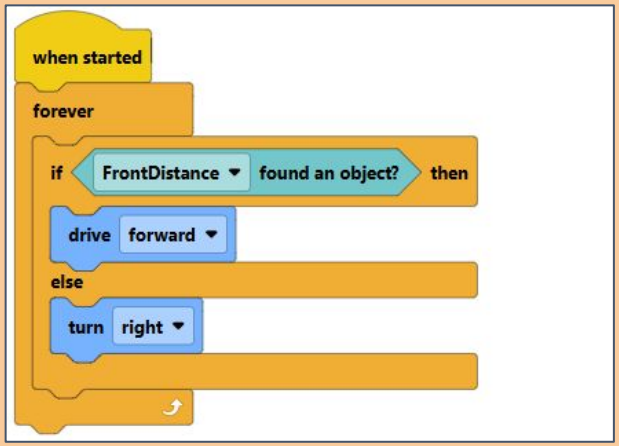


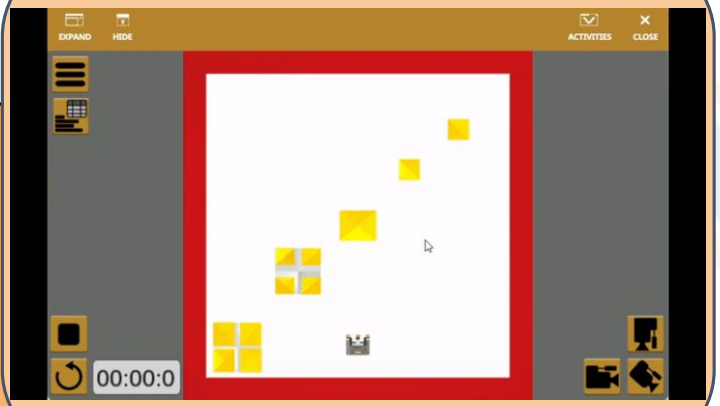
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The Attack

Lets see the action happen on the robot!





1 The Attack

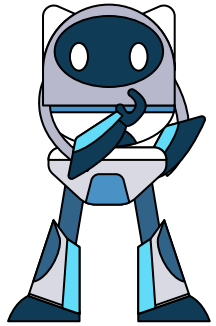
- Now after scanning, we need the robot to do something.
- So, we put 'drive forward' block under 'if' section.

2 Run Result

- Now we can see the drive onto the castle.
- It will also turn right until it see the object and keep driving forward.
- But this cause the robot go beyond red line and fall down with the castle.

The method is working but it fail to knock all the castle.

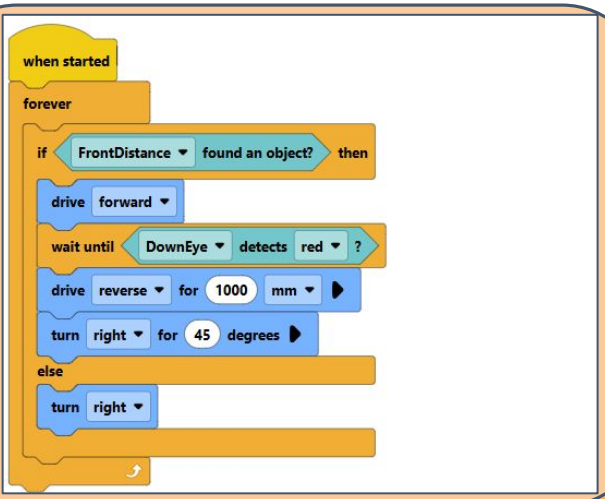
Do you know how do we fix this?

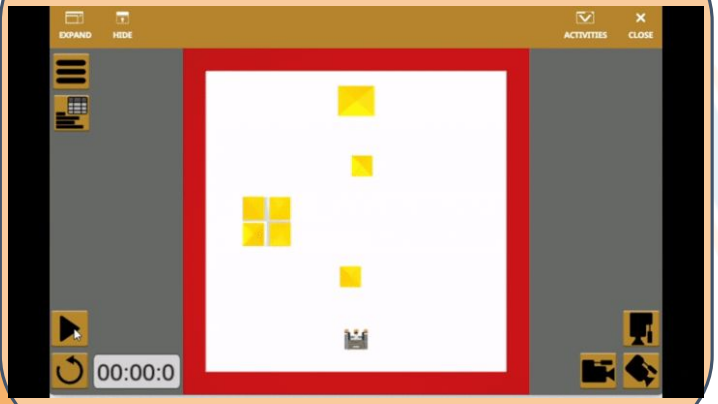


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Safety Switch

There is always a fix for the problem.





1 Safety switch

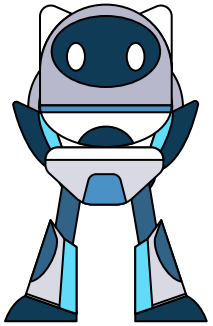
- To fix the falling problem, we need to add a safety switch for robot to stop before falling.
- So, we add 'wait until' and inside is 'DownEye detects red'.
- Then we put 'drive reverse' and 'turn right' after it detect the red colour.

2 Run Result

- Now we can see whenever the robot detect red line, it will reverse and turn until it fine another castle.
- This will prevent the robot to fall down and can search all the castle to knock it down.

Great! Now it is working properly as intended.

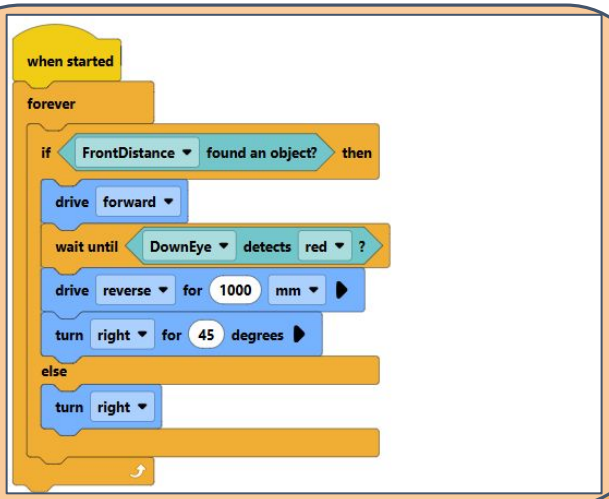
Feel free to be creative and adjust the code to achieve the same objective!

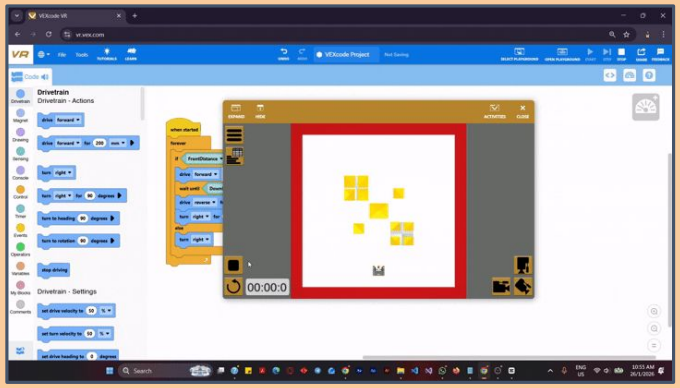


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Conclusion

Congrats on your second project!





1 Code Review

- With our current code, we learn that every code need a safety switch for any possibilities that can happen.
- We use a nested loop where we are using loop inside a loop with 'forever' block and 'wait until' block.

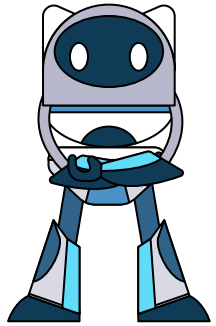
2 Run Result

- In any castle placement, we still manage to knock all of them.
- Do you realize we manage to make a self-driving cars?
- With this as a base for learning, more advance invention can be make in future too.

Remember, every big project start from small base learning.

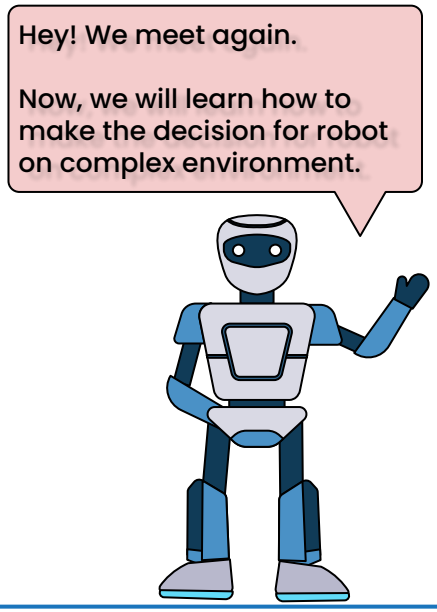
Learn well and lets be prepared on next project.

Good Luck!



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The Smart Mouse



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Wall Maze Playground

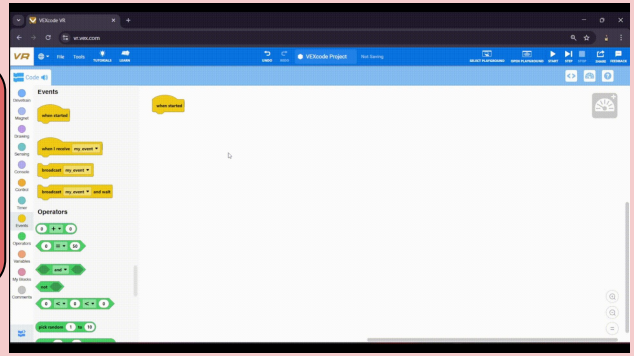
Navigating the robot through layer of code execution.



Wall Maze

1 What is maze?

- Maze is a playground where there is multiple choice of path to navigate.
- Every single path need a decision to ensure the robot can reach to destination.
- How can a robot make a decision that has a lot of choices?

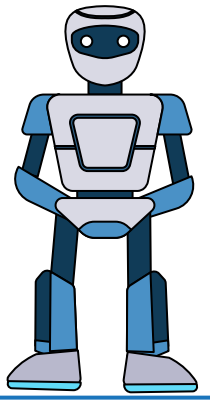


2 Setup

- Now let's open the maze.
- Press 'select playground' button on top right of the screen.
- Then, scroll down and choose wall maze.

Maze is one of famous problem on figuring out path finding for robot.

As maze has complex path, it challenge the robot on making lot of decision in a single run.

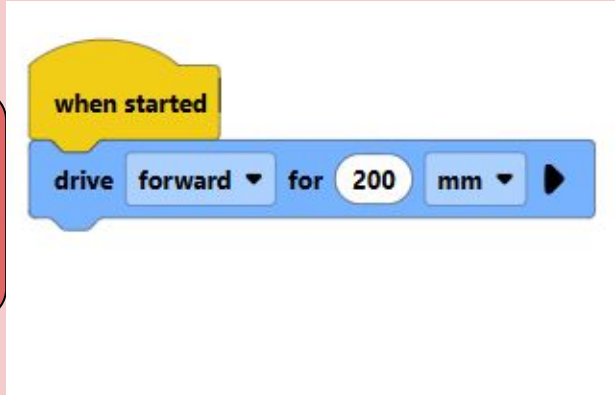


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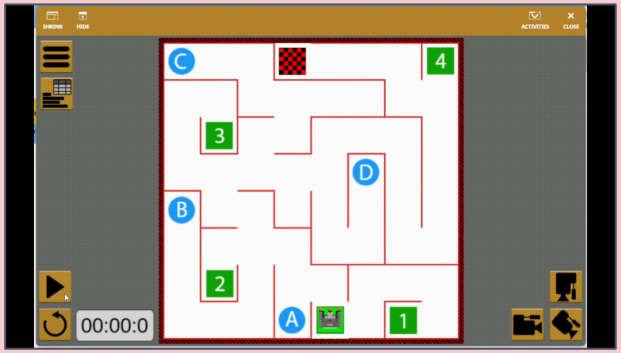
The Blind Robot

Demonstrating the problem on hard coding.



1 Planning the code

- Alright, Lets test with simple 'drive forward for 200mm' block.
- Can we simply go through the maze and reach the checkerboard as our objective?

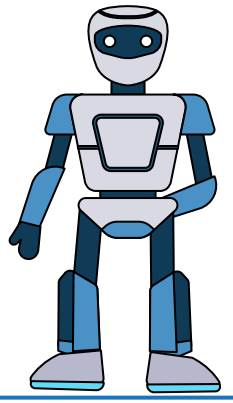


2 Run Result

- From the code, the robot follow exactly as our code.
- We could hardcoded and put one by one turn and drive forward.
- But the code will get too long and cannot work on different maze.

How much time it will take for us to put code one by one till we reach checkerboard?

It is easy for the first maze but what if the maze get too long?

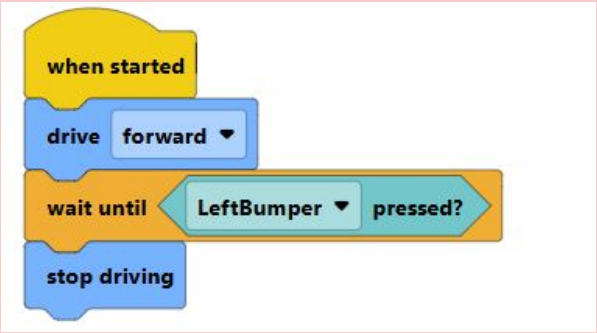


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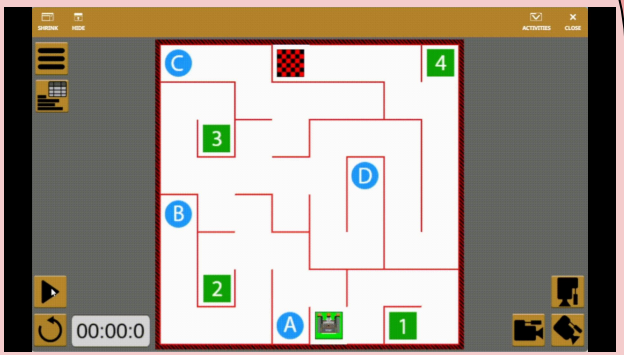
Coding the sense

For our robot to make decision on touch.



1 Code Adjustment

- Now, lets adjust the code a little.
- We will just put ‘drive forward’ for the robot to keep moving.
- ‘Wait until leftbumper pressed?’ for the code to execute next only when the left bumper hit anything.
- If the robot’s left bumper hit, the robot will stop moving.

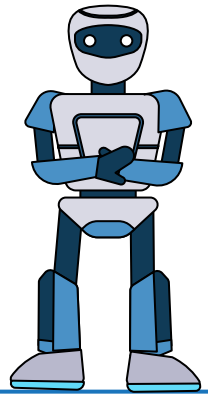


2 Run Result

- Now, the robot can sense there is wall infront and stop driving.
- But we are not done. Lets keep moving onto next step.

Remember!

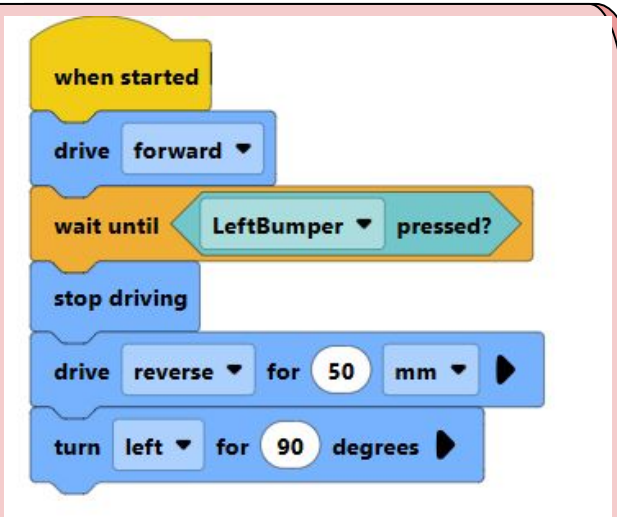
You can also check on sensor monitor to see how the sensor bumper work.



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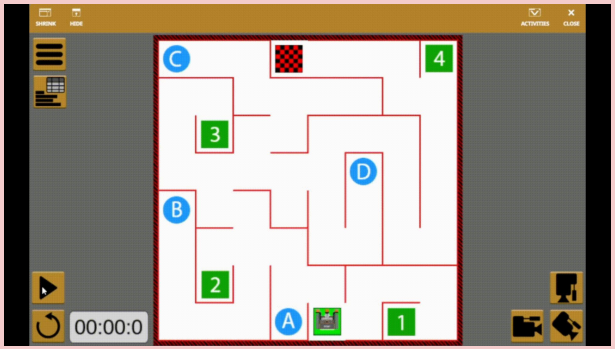
Reaction on touch

The robot need to have some action when condition are meet.



1 Code Adjustment

- We manage to make the robot sense and make it stop before.
- Now, lets put 'reverse for 50mm' so our robot can reposition after touching the wall.
- Then, put 'turn left for 90 degrees' block.
- Lets see the result.

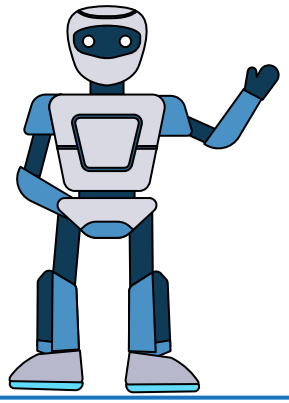


2 Run Result

- The robot manage to go back and turn left at a good position.
- If the robot somehow is not in good position, do adjust the value according to the code.

Anyone has problem so far?

Raise a hand if you need help!

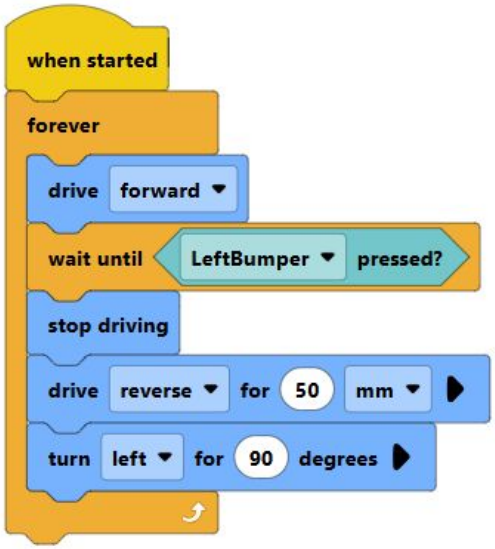


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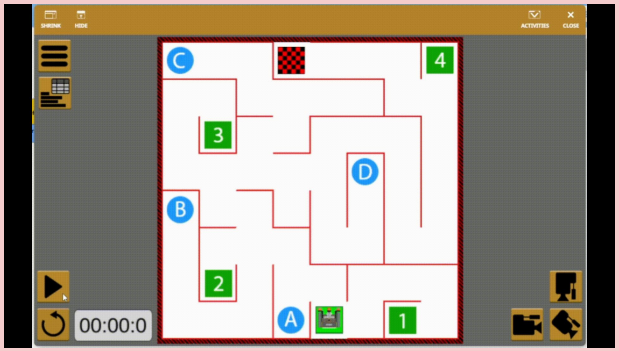
The Loop

Its time to automate the movement for the robot!



1 Code Adjustment

- Now lets wrap our code with 'forever' block.
- This will make the robot keep executing the code inside forever.



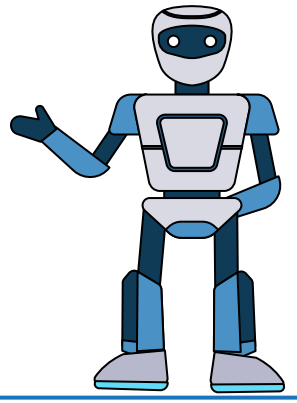
2 Run Result

- WIth this, the robot manage to reach to platform A and 2.
- But, it fail to get out from platform 2.
- Reason is, Block 2 are surrounded with wall with narrow space on out.

This is the loop space for our robot.

Which mean, our robot cannot go out with the current code.

How do we improve it?

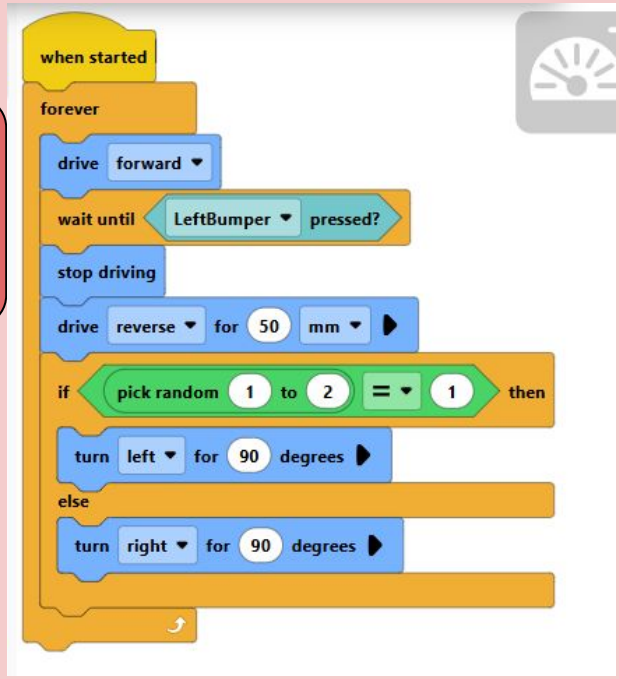


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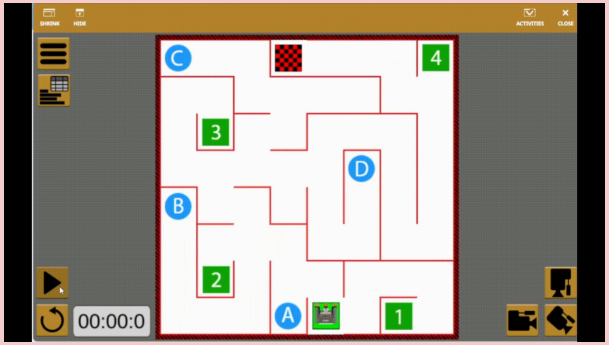
Optimization

For robot to not get stuck and drive better.



1 Code Adjustment

- Alright now, try to adjust the code same as image above.
- This code use pick random for a toss coin, for random path chosen for robot to go left or right.



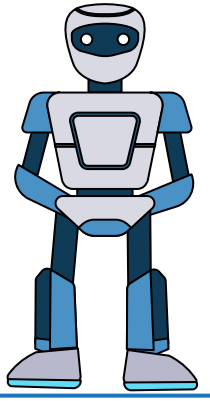
2 Run Result

- With the new code, robot now move to left or right with luck.
- This help the robot to finally get out from platform 2 after several rerun.

Now, we are running by through coin flip luck.

Eventually we will breakthrough the problem and manage to get out.

But, is it a good or efficient way?

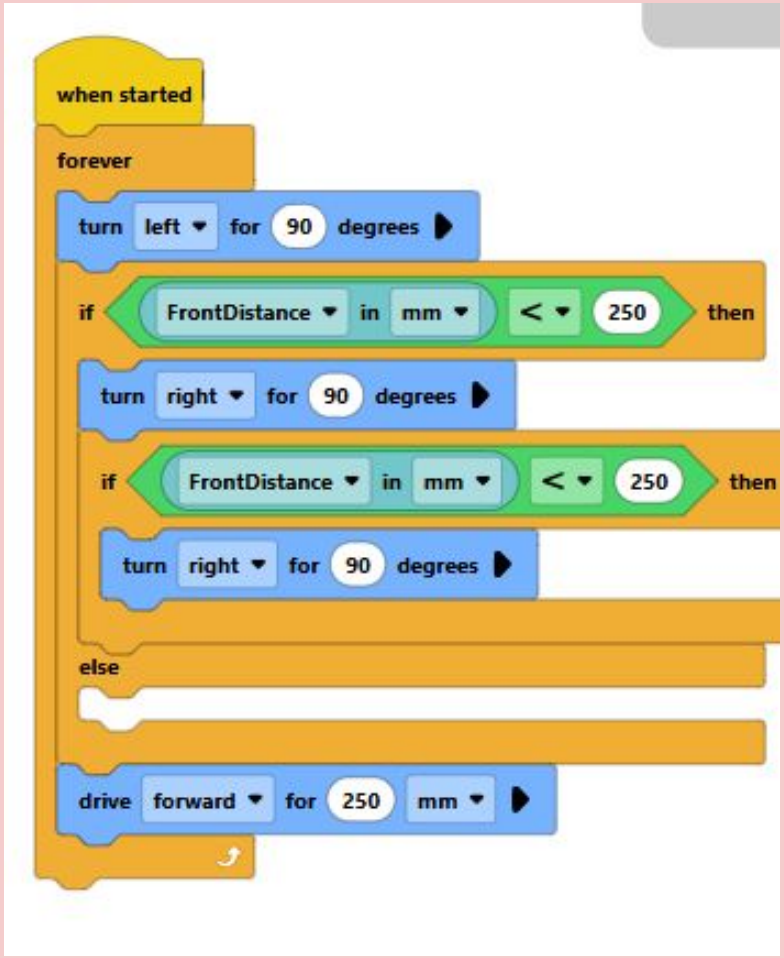


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Left Hand Rule

There is always solution on every problem



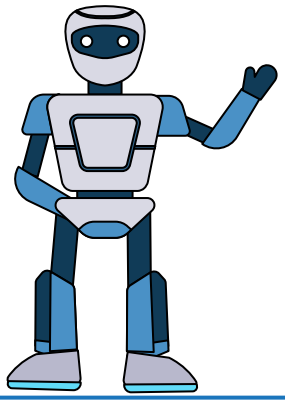
Code Adjustment

- Do you remember our primary objective?
- Now while we do manage to unstuck the car, how do we get to the checkerboard?
- Thus on the left, is the code to execute Left Hand Rule.
- Left hand rule are one of practice on escaping the maze by following the leftside of the wall.
- The robot will always turn left on intersection.
- Upon meeting wall with sensor (Frontdistance in mm), it will turn right first and keep turning right until it doesnt met any wall.

Making a lot of decision are hard with only some code.

But efficient code can make it possible .

Now, we will learn on how to make efficient code.



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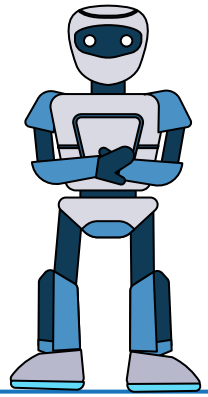
Escaping the maze.

Can we escape the maze with left-hand rule?



Now, we can see it check row by row on wall.

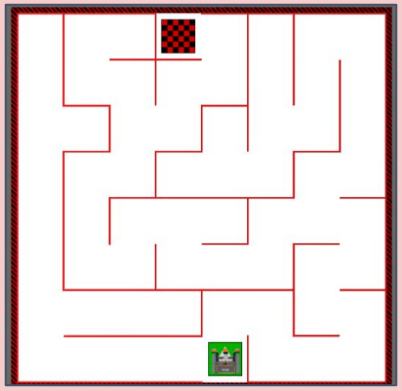
With this, it can always find a way to escape any maze although it takes time to complete.

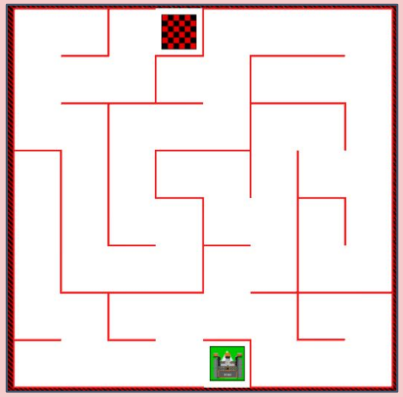


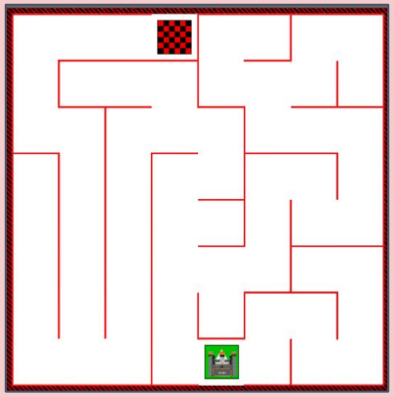
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Conclusion

Finishing the 3rd project in Vex Code VR!





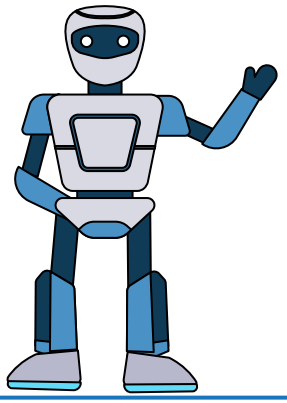


- Congratulations on finishing your smart mouse project.
- Do you know what this project remind of? A Roomba Vacuum Cleaner!
- A roomba will clean the floor and if it bump any object, it will reverse and go to another path.
- Same as real life, any house layout are not the same.
- Hence we develop a way for our robot suitable for any house layout by training it in a maze.
- Keep learning coder warrior, there is always a room for improvement on our code.

Congrats coder. It was tough but you manage to get here.

Remember in code life, there is always new knowledge we can learn.

Learn, Improvise, Adapt.

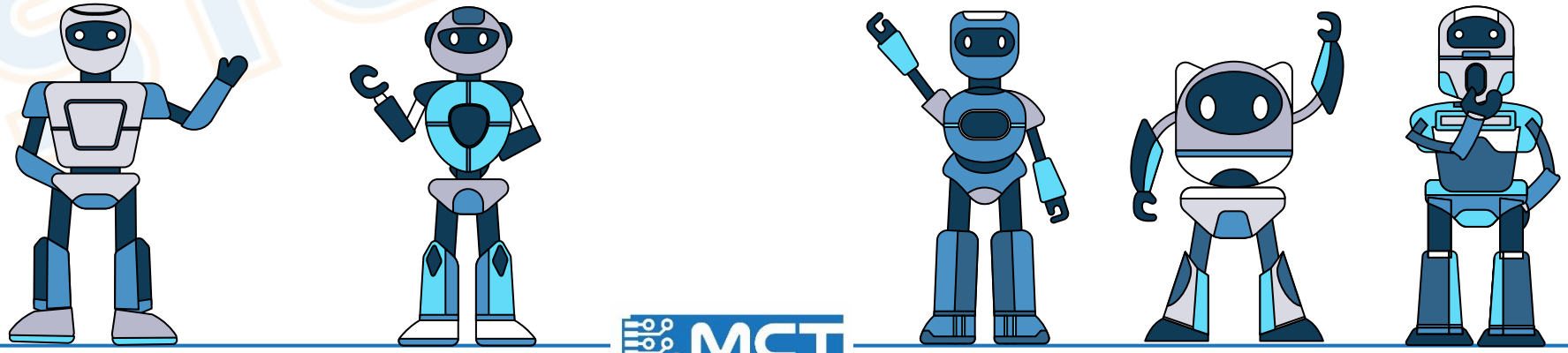


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Thank You!

Make it work, make it right, make it fast.

(Kent Beck)



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